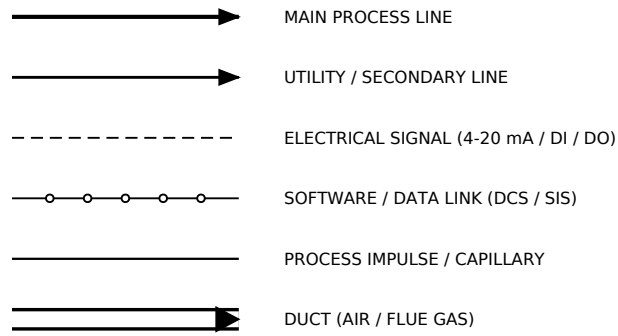
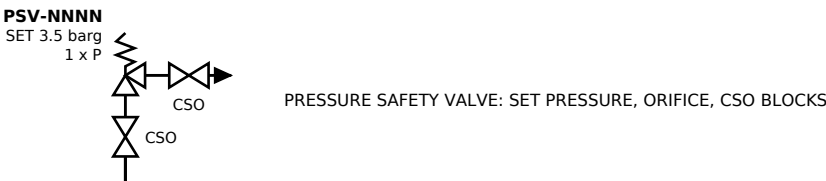
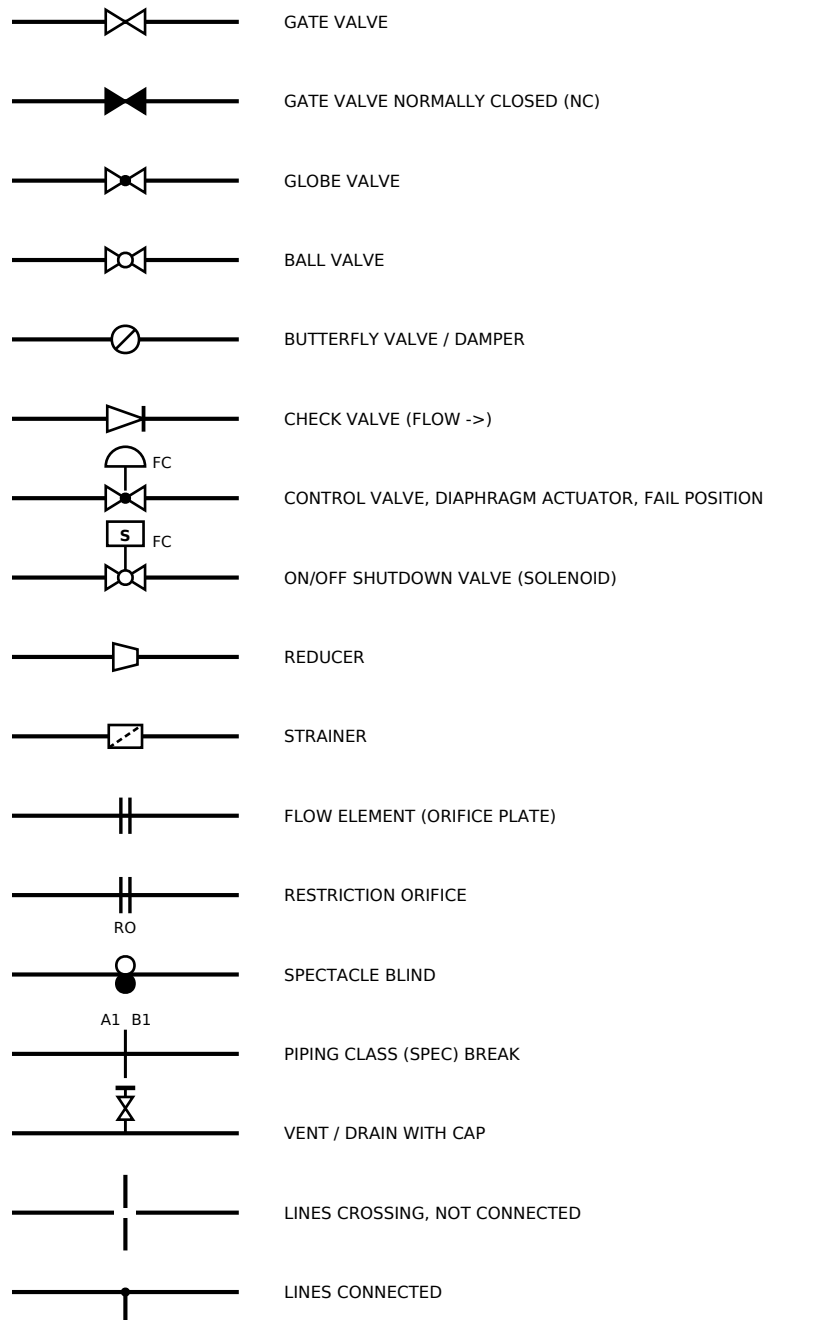


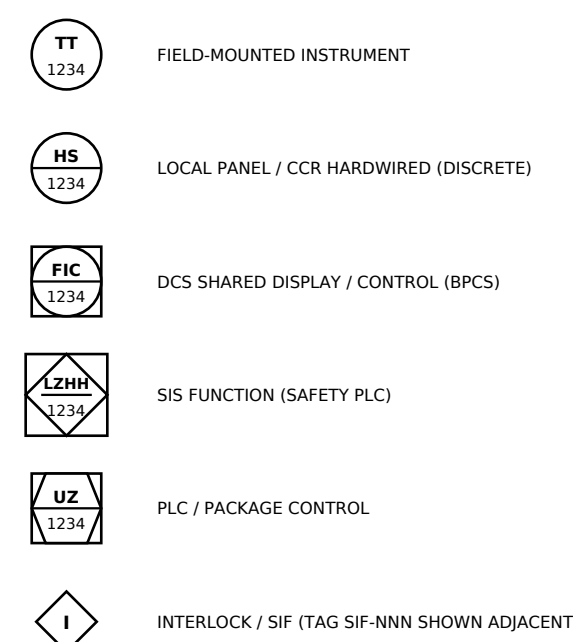
LINE SYMBOLS



VALVES & PIPING COMPONENTS



INSTRUMENT SYMBOLS (ISA-5.1)



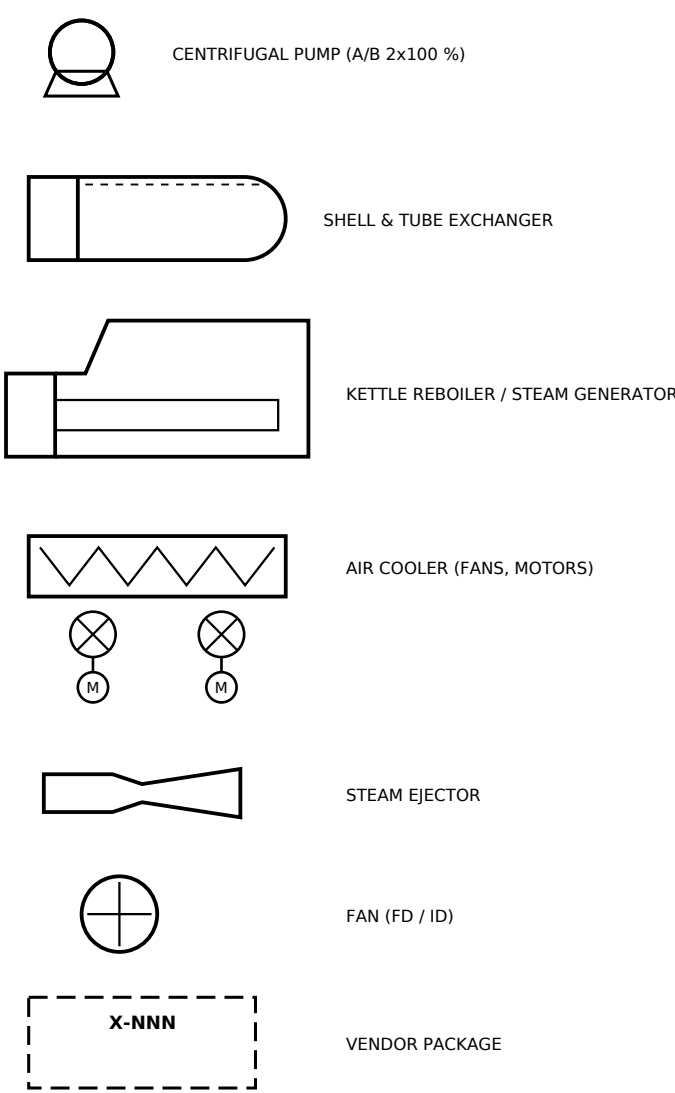
INSTRUMENT IDENTIFICATION LETTERS

A ANALYSIS (O₂, pH) B BURNER / FLAME
F FLOW H HAND
L LEVEL P PRESSURE
T TEMPERATURE X/Z ON-OFF / POSITION
PD / TD DIFFERENTIAL FF FLOW RATIO
Succeeding: E element, G glass, I indicate, C control,
T transmit, V valve, Y relay/compute, S switch, Z (2x4) SIS,
AH/AH/HI/ALL alarms; A/B/C suffix = redundant SIS

ABBREVIATIONS

FC / FO / FL FAIL CLOSED / OPEN / LAST
CSO / CSC CAR-SEALED OPEN / CLOSED
NC / NO NORMALLY CLOSED / OPEN
NMF NORMALLY NO FLOW
TPA / MPA / BPA TOP / MID / BOTTOM PUMPAROUND
SS STRIPPING STEAM
DBB DOUBLE BLOCK & BLEED
ROSOV / EIV REMOTE OPERATED SHUT-OFF VALVE
COI / COT COIL OUTLET / INLET TEMP.
RO RESTRICTION ORIFICE
FA FLAME ARRESTOR
OSBL OUTSIDE BATTERY LIMITS
BMS BURNER MANAGEMENT SYSTEM
2o3 TWO-OUT-OF-THREE VOTING

EQUIPMENT SYMBOLS



LINE NUMBER

12"-P-100-002-B1-H

NPS (in) - FLUID - AREA - SEQUENCE - PIPING CLASS - INSULATION
AREA: 100 CDU, 200 VDU, 900 UNIT UTILITIES

FLUID CODES

P Process hydrocarbon
PG Process gas / vapour
SW Sour water
WW Wash water / brine
LS LP steam
HS MP steam
HS HP steam
CD Condensate
BFW Boiler feed water

CWS Cooling water supply
CWR Cooling water return
FG Fuel gas
FL Flare
BD Blowdown / drain
IA Instrument air
N Nitrogen
CH Chemical

INSULATION CODES

H HEAT CONSERVATION
P PERSONNEL PROTECTION
ST STEAM TRACED + INSULATED
C COLD
N NONE

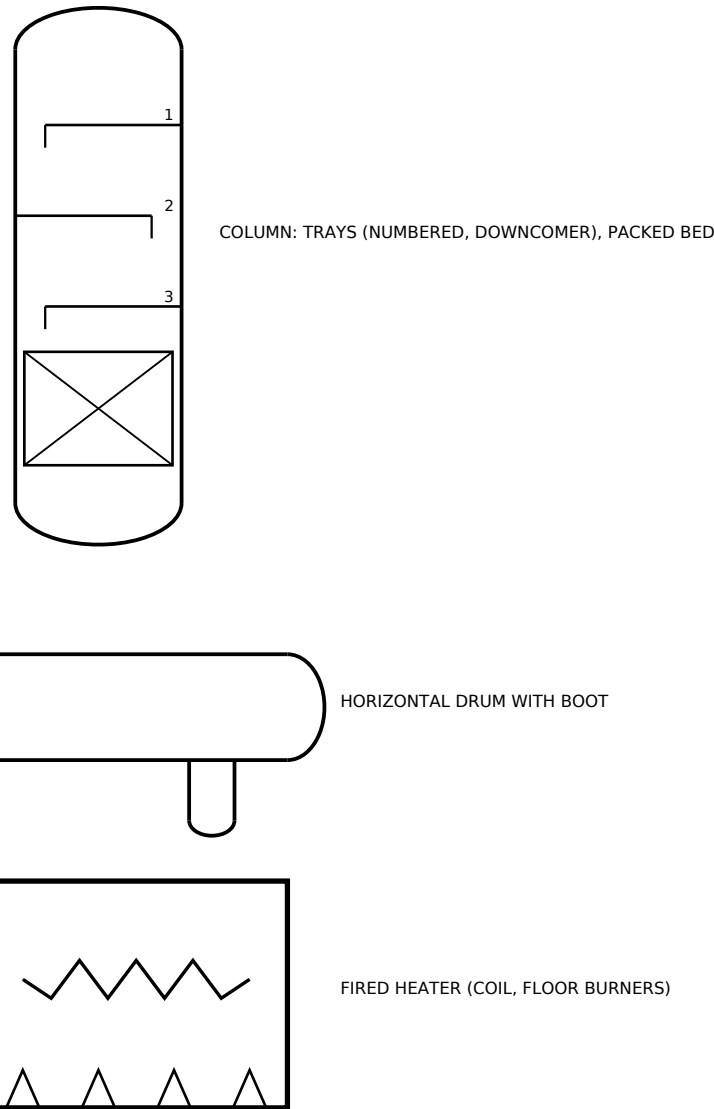
PIPING CLASSES (SUMMARY, SEE CFU-000-PI-SPC-001)

A1 150# Carbon steel ASTM A106 Gr.B
A2 150# Carbon steel, HIC-resistant (NACE MR0103)
B1 300# Carbon steel ASTM A106 Gr.B
B2 300# SC-1/2Mo ASTM A335 P5
B3 300# 9Cr-1Mo ASTM A335 P9
C1 600# Carbon steel, killed ASTM A106 Gr.B
A3 300# Carbon steel, HIC-resistant (NACE MR0103)
S1 150# Carbon steel ASTM A106 Gr.B
S2 600# 1.25Cr-1/2Mo ASTM A335 P11
S3 300# Carbon steel ASTM A106 Gr.B
F1 150# Carbon steel, killed, impact-tested (A33)
U1 150# Carbon steel (cement-lined for CW >= 12°)
U2 150# SS304 (A) / galvanised CS (N2)
DRAWING INDEX A335 P5

Hydrocarbon <= 230 C, general
Sour water, OH vapour/condensate, wash water/brine
Hydrocarbon <= 260 C, pump discharges
Sulfidic hydrocarbon 260-400 C (incl crude, AR, BPA, HvGO, VR)
Heater outlets / transfer lines (H-101, H-201) and > 400 C
LPG / stabiliser, BFW > 300#
Wash water / brine / sour water above 150# rating (desalter pr
LP / MP steam, condensate
HP steam 41 barg / 400 C
MP steam 10.3 barg / 250-290 C design (exceeds S1)
Flare / relief headers, -29 to 350 C
Cooling water, utility water, LP BFW
Instrument air, nitrogen
Vacuum vapour lines (C-201 OH); transfer line > 400 C -> B3

LEGEND, SYMBOLS & ABBREVIATIONS
CRUDE CHARGE PUMPS & COLD PREHEAT TRAIN
ELECTROSTATIC DESALTERS D-101A/B & BOOSTER PUMPS
HOT PREHEAT TRAIN E-106 TO E-111 & E-113
ATMOSPHERIC HEATER H-101 - PROCESS COILS
ATMOSPHERIC HEATER H-101 - FIRING & AIR SYSTEM
ATMOSPHERIC COLUMN C-101 - FLASH ZONE & BOTTOMS
ATMOSPHERIC COLUMN C-101 - UPPER SECTION & PUMPAROUNDS
SIDE STRIPPERS C-102/103/104 & PRODUCT RUNDOWN
C-101 OVERHEAD SYSTEM & REFLUX DRUM D-102
NAPHTHA STABILISER C-105
NAPHTHA SPLITTER C-106
VACUUM HEATER H-201
VACUUM COLUMN C-201
VACUUM PUMPAROUNDS & PRODUCTS
VACUUM EJECTOR SYSTEM & HOTWELL D-201
FUEL GAS, FLARE KO & UTILITY HEADERS

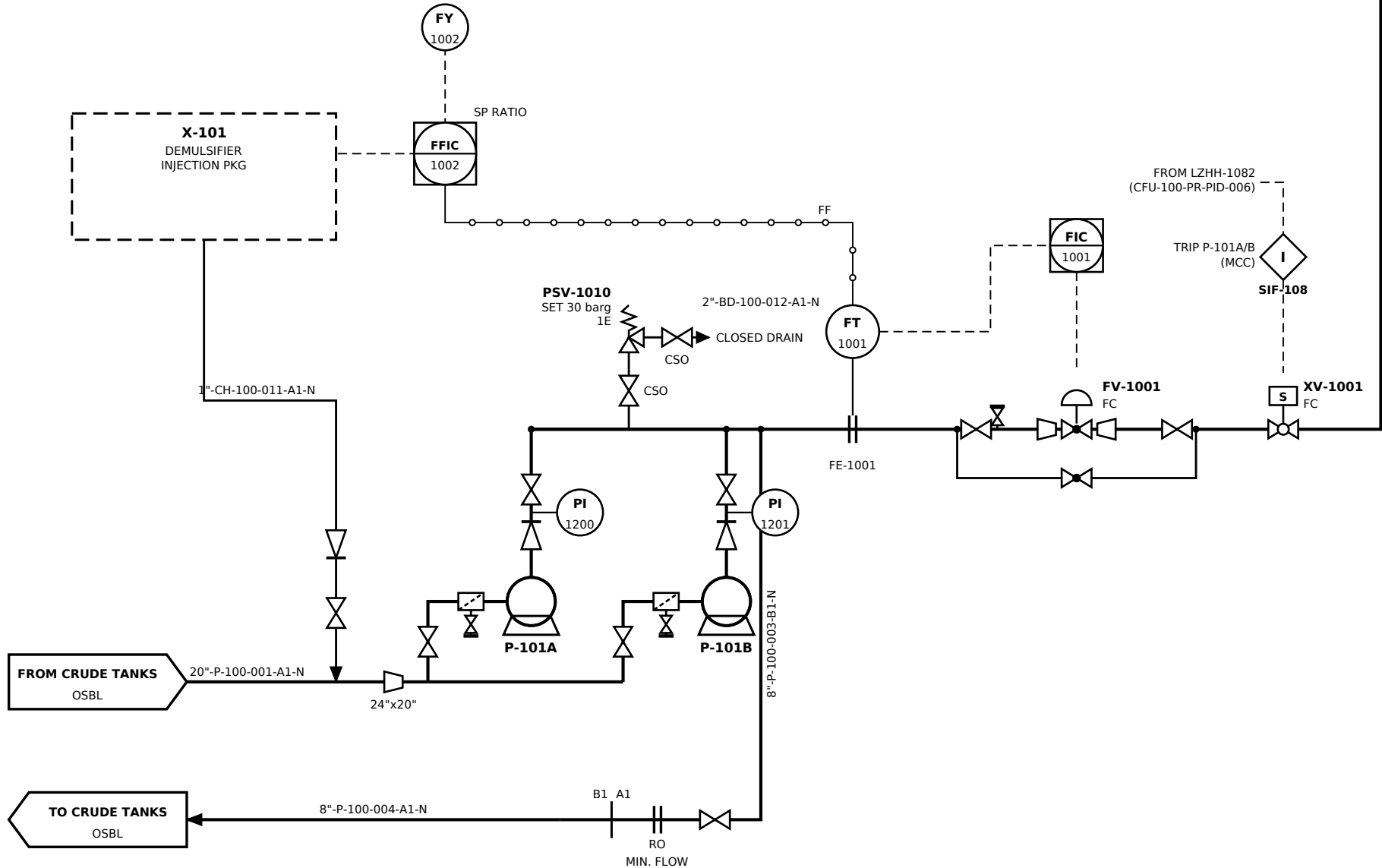
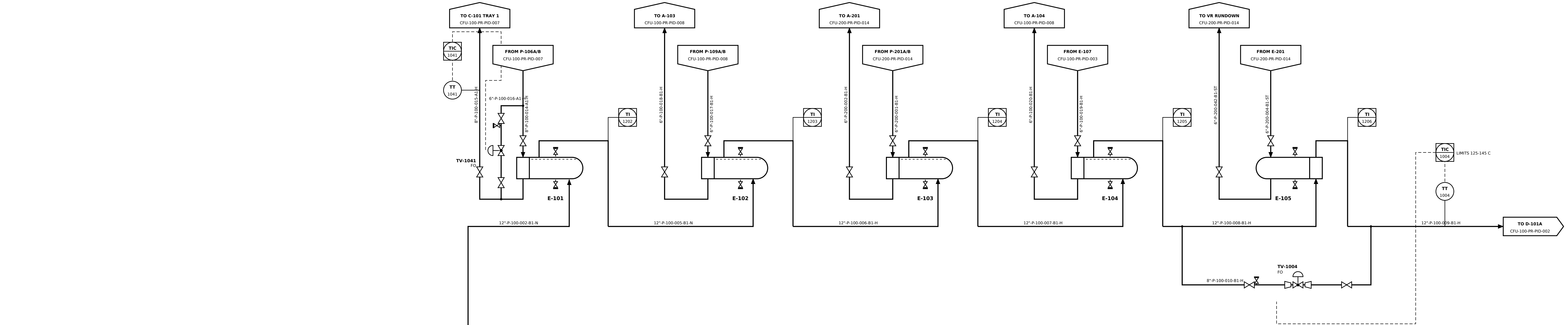
EQUIPMENT DATA BOX



NOTES
1. Legend applies to all P6&Ds CFU-xxx-PR-PID-001 to -016.

REV	DATE	DESCRIPTION	
A	2026-10-02	Issued for review (FEED)	
GENERIC REFINERY - GULF COAST 100 KBPSD CRUDE & VACUUM DISTILLATION UNIT			
PIPING & INSTRUMENT DIAGRAM LEGEND, SYMBOLS & ABBREVIATIONS			
DISCIPLINE	STAGE	SCALE	SHEET
PROCESS	FEED	NTS	1 OF 17
DRAWN	CHECKED	APPROVED	DATE
CLAUDE CODE	-	-	2026-10-02
DRAWING NUMBER			REV
CFU-000-PR-PID-000			A

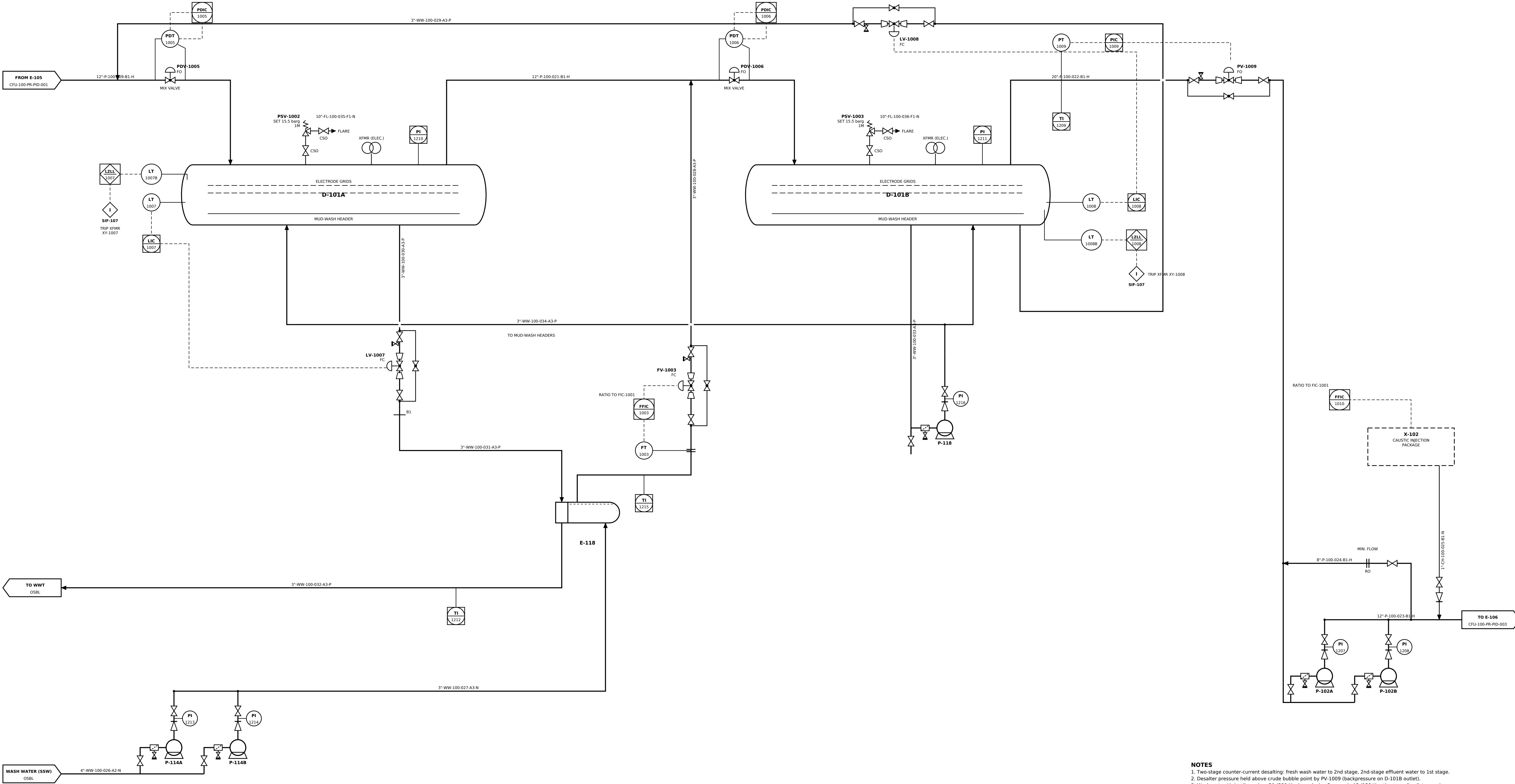
P-101A/B Crude charge RATED 730 m3/h @ 261 m, @P 22 bar MOTOR 710 kW, API 682, Flan 11/22 DES. T 60 C; MAT'L S-6 (CS/12Cr)	E-101 Crude / Top pumparound 359 m2 total; 15 x 1P shells DESIGN 35 / 20 barg / 185 C; CA 3 mm DUTY 8.09 MW SHELL: Crude / TUBE: Top pumparound MAT'L CS shell / CS tubes	E-102 Crude / Kerosene product 438 m2 total; 15 x 1P shells DESIGN 35 / 20 barg / 240 C; CA 3 mm DUTY 6.55 MW SHELL: Crude / TUBE: Kerosene product MAT'L CS shell / CS tubes	E-103 Crude / LVGO pumparound/product 804 m2 total; 25 x 1P shells DESIGN 35 / 20 barg / 255 C; CA 3 mm DUTY 11.45 MW SHELL: Crude / TUBE: LVGO pumparound/product MAT'L CS shell / CS tubes	E-104 Crude / Diesel product 378 m2 total; 15 x 1P shells DESIGN 35 / 20 barg / 235 C; CA 3 mm DUTY 4.44 MW SHELL: Crude / TUBE: Diesel product MAT'L CS shell / CS tubes	E-105 Crude / Vacuum residue 431 m2 total; 15 x 1P shells DESIGN 35 / 20 barg / 285 C; CA 3 mm DUTY 5.84 MW SHELL: Vacuum residue / TUBE: Crude MAT'L CS shell / 5Cr-1/2Mo tubes	X-101 Demulsifier injection package Tank 2 m3 + 2 x 100 % metering pumps MOTOR 0.75 kW MAT'L SS316
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- NOTES**
1. Cold-train crude-side design 35 barg (E-101, E-105) >= P-101 shut-off; PSV-1010 (30 barg) thermal / blocked-in.
 2. Exchanger hot-side connections shown at channel / shell nozzles; TEMA type & shells per data sheets.
 3. TV-1041 / TV-1004 bypass control valves have block valves only (exchanger provides bypass path).
 4. Demulsifier dosed at P-101 suction, ratio to FIC-1001 (FFIC-1002) - X-101 vendor package.
 5. Line numbers: <NPS>-<fluid>-<area>-<seq>-<class>-<insulation> per CONVENTIONS; see line list CFU-000-PI-LL-001.
 6. All vents / drains 3/4" min. with blind or cap; HPV/LPD to piping specification. Valve sizes = line size UNO.
 7. Instrument tags per ISA-5.1; index CFU-000-IC-IDX-001. SIS functions per SRS (SIF tags shown in interlocks).

REV	DATE	DESCRIPTION	
A	2026-10-02	Issued for review (FEED)	
GENERIC REFINERY - GULF COAST			
100 KBPSD CRUDE & VACUUM DISTILLATION UNIT			
PIPING & INSTRUMENT DIAGRAM			
CRUDE CHARGE PUMPS & COLD PREHEAT TRAIN			
DISCIPLINE	STAGE	SCALE	SHEET
PROCESS	FEED	NTS	2 OF 17
DRAWN	CHECKED	APPROVED	DATE
CLAUDE CODE	-	-	2026-10-02
DRAWING NUMBER			REV
CFU-100-PR-PID-001			A

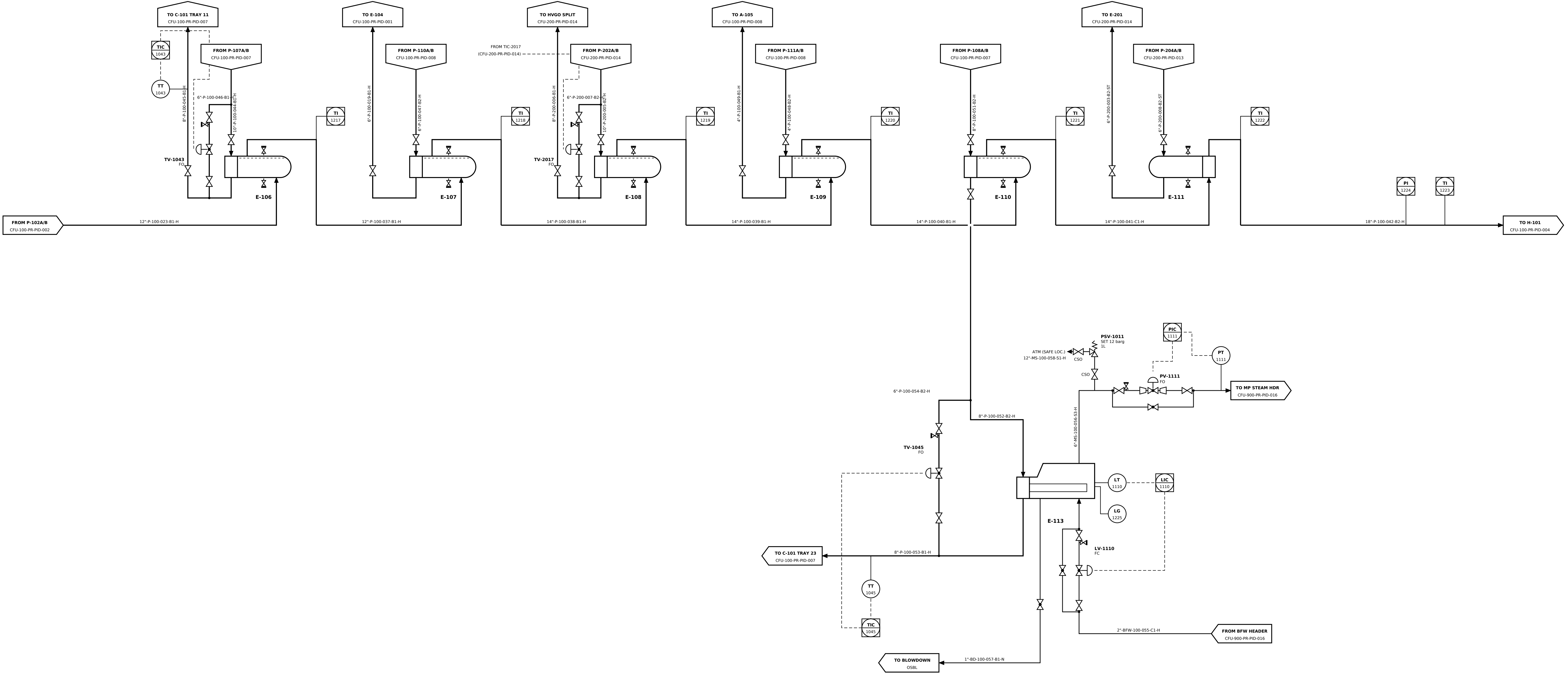
D-101A Electrostatic desalter, 1st stage ID 3.8 m x 30.0 m TTT, ACDC dual-polarity grids DESIGN 17.0 barg / 165 C, CA 3 mm MAT'L CS	D-101B Electrostatic desalter, 2nd stage ID 3.8 m x 30.0 m TTT, ACDC dual-polarity grids DESIGN 17.0 barg / 165 C, CA 3 mm MAT'L CS	E-118 Desalter wash water / brine 84 m2, 1 shells DESIGN 20 / 20 barg / 160 C, CA 3 mm DUTY 1.62 MW SHELL: Wash water / TUBE: Brine MAT'L CS shell / duplex 2205 tubes	P-102A/B Desalted crude booster RATED 805 m3/h @ 263 m, dP 20 bar MOTOR 713 kW, API 682, Plan 11052 DES. T 165 C, MAT'L S-6 (CS/12Cr)	P-114A/B Desalter wash water RATED 37 m3/h @ 144 m, dP 14 bar MOTOR 30 kW, API 682, Plan 11052 DES. T 80 C, MAT'L S-6 (CS/12Cr)	P-118 Desalter mud-wash / recycle RATED 19 m3/h @ 64 m, dP 6 bar MOTOR 7.5 kW, API 682, Plan 11052 DES. T 150 C, MAT'L S-6 (CS/12Cr)	X-102 Caustic injection package (desalted crude) Tank 2 m3 + 2 x 100 % metering pumps MOTOR 0.75 kW MAT'L SS316
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- NOTES**
- Two-stage counter-current desalting; fresh wash water to 2nd stage, 2nd-stage effluent water to 1st stage.
 - Desalter pressure held above crude bubble point by PV-1009 (backpressure on D-101B outlet).
 - Wash water / brine lines class B1 (300#) where design P exceeds A2 (150#) rating - see line list remarks.
 - Transformer trip on low-low interface (SIF-107) by Electrical (XY-1007 / XY-1008 to feeder CB).
 - Line numbers: <NPS>-<fluid>-<area>-<seq>-<class>-<insulation> per CONVENTIONS; see line list CFU-000-PI-LL-001.
 - All vents / drains 3/4" min. with blind or cap; HPV/LPD to piping specification. Valve sizes = line size UNO.
 - Instrument tags per ISA-5.1; index CFU-000-IC-IDX-001. SIS functions per SRS (SIF tags shown in interlocks).

REV	DATE	DESCRIPTION	
A	2026-10-02	Issued for review (FEED)	
GENERIC REFINERY - GULF COAST 100 KBPSD CRUDE & VACUUM DISTILLATION UNIT			
PIPING & INSTRUMENT DIAGRAM ELECTROSTATIC DESALTERS D-101A/B & BOOSTER PUMPS			
DISCIPLINE	STAGE	SCALE	SHEET
PROCESS	FEED	NTS	3 OF 17
DRAWN	CHECKED	APPROVED	DATE
CLAUDE CODE	-	-	2026-10-02
DRAWING NUMBER			REV
CFU-100-PR-PID-002			A

E-106 Crude / Middle (kero) pumparound 1001 m2 total; 25 x 1P shells DESIGN 45 / 20 barg / 270 C; CA 3 mm DUTY 5.83 MW SHELL: Crude / TUBE: Middle (kero) pumparound MATL:CS shell / 5Cr-1/2Mo tubes	E-107 Crude / Diesel product 324 m2 total; 15 x 1P shells DESIGN 45 / 20 barg / 325 C; CA 6 mm DUTY 5.83 MW SHELL: Crude / TUBE: Diesel product MATL:CS shell / 5Cr-1/2Mo tubes	E-108 Crude / HVGO pumparound/product 2196 m2 total; 25 x 2P shells DESIGN 45 / 20 barg / 345 C; CA 6 mm DUTY 21.14 MW SHELL: Crude / TUBE: HVGO pumparound/product MATL:1.25Cr shell / 9Cr-1Mo tubes	E-109 Crude / AGO product 263 m2 total; 15 x 1P shells DESIGN 45 / 20 barg / 345 C; CA 6 mm DUTY 3.11 MW SHELL: Crude / TUBE: AGO product MATL:1.25Cr shell / 9Cr-1Mo tubes	E-110 Crude / Bottom (diesel) pumparound 886 m2 total; 15 x 2P shells DESIGN 45 / 20 barg / 345 C; CA 6 mm DUTY 7.58 MW SHELL: Crude / TUBE: Bottom (diesel) pumparound MATL:1.25Cr shell / 9Cr-1Mo tubes	E-111 Crude / Vacuum residue 1094 m2 total; 15 x 2P shells DESIGN 45 / 20 barg / 390 C; CA 6 mm DUTY 7.58 MW SHELL: Vacuum residue / TUBE: Crude MATL:1.25Cr shell / 9Cr-1Mo tubes	E-113 BPA / MP steam generator (kettle) 223 m2; 1 shell(s) DESIGN 13 / 20 barg / 290 C; CA 3 mm DUTY 4.91 MW SHELL: BFW / MP steam / TUBE: Bottom pumparound MATL:CS
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- NOTES**
- Hot-train crude-side design 45 barg (E-106, E-111) >= P-102 shut-off (D-1018 design + 1.2 x dPI).
 - E-113 kettle MP steam generator: 3-element level control simplified to LIC-1110 at FEED.
 - TV-2017 controlled from TIC-2017 (HVGO pumparound return temperature) on CFU-200-PR-PID-014.
 - Line numbers: <NPS>-<fluid>-<area>-<seq>-<class>-<insulation> per CONVENTIONS; see line list CFU-000-PI-LL-001.
 - All vents / drains 3/4" min. with blind or cap; HPV/LPD to piping specification. Valve sizes = line size UNO.
 - Instrument tags per ISA-5.1; index CFU-000-IC-IDX-001. SIS functions per SRS (SIF tags shown in interlocks).

REV	DATE	DESCRIPTION	
A	2026-10-02	Issued for review (FEED)	
GENERIC REFINERY - GULF COAST 100 KBPSD CRUDE & VACUUM DISTILLATION UNIT			
PIPING & INSTRUMENT DIAGRAM HOT PREHEAT TRAIN E-106 TO E-111 & E-113			
DISCIPLINE		STAGE	SHEET
PROCESS		FEED	NTS 4 OF 17
DRAWN	CHECKED	APPROVED	DATE
CLAUDE CODE	-	-	2026-10-02
DRAWING NUMBER			REV
CFU-100-PR-PID-003			A

H-101
Atmospheric crude charge heater
60.7 MW abs. / 67.4 MW fired; 2-cell cabin, 8 passes, 144 rad. tubes 6" x 18.3 m
DESIGN 38.5 barg / 545 C; CA 3 mm
DUTY 60.68 MW
MAT'L Tubes A335 P9 (9Cr-1Mo); convection P5 / CS studded

LP STEAM HEADER
CFU-900-PR-PID-016

FROM E-111 (CIT)
CFU-100-PR-PID-003

BFW HEADER
CFU-900-PR-PID-016
1" BFW-100-072-016-H

TV-1024
FC

TIC
1024

350 C
TT
1024

PI
1227

STRIPPING STEAM HDR
CFU-100-PR-PID-006

TT
1020A-C

2003
TSW
1025

SIF-106

TO C-101 FLASH ZONE
CFU-100-PR-PID-006

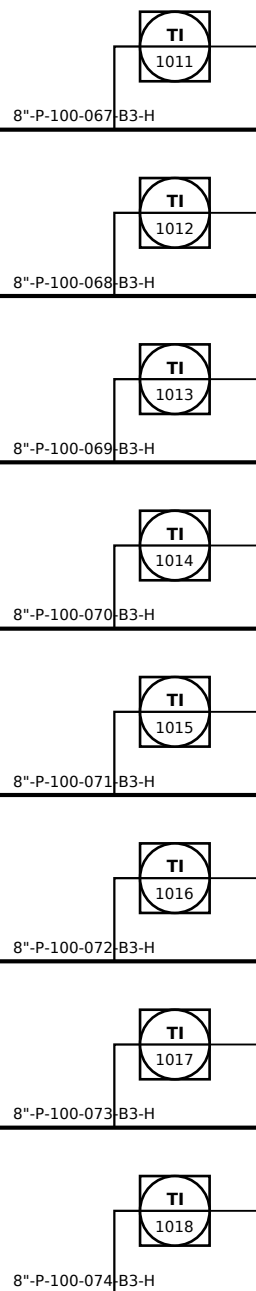
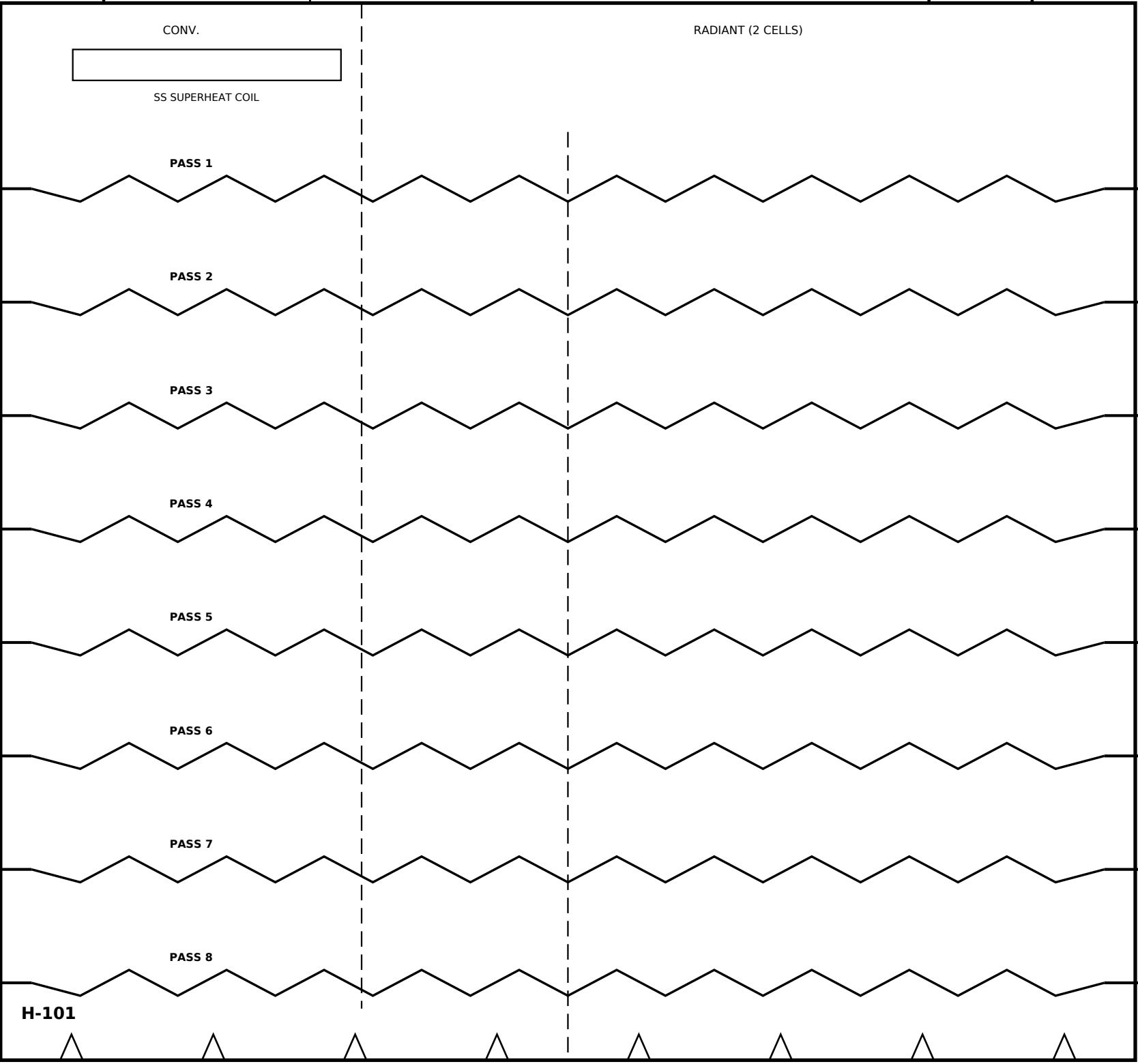
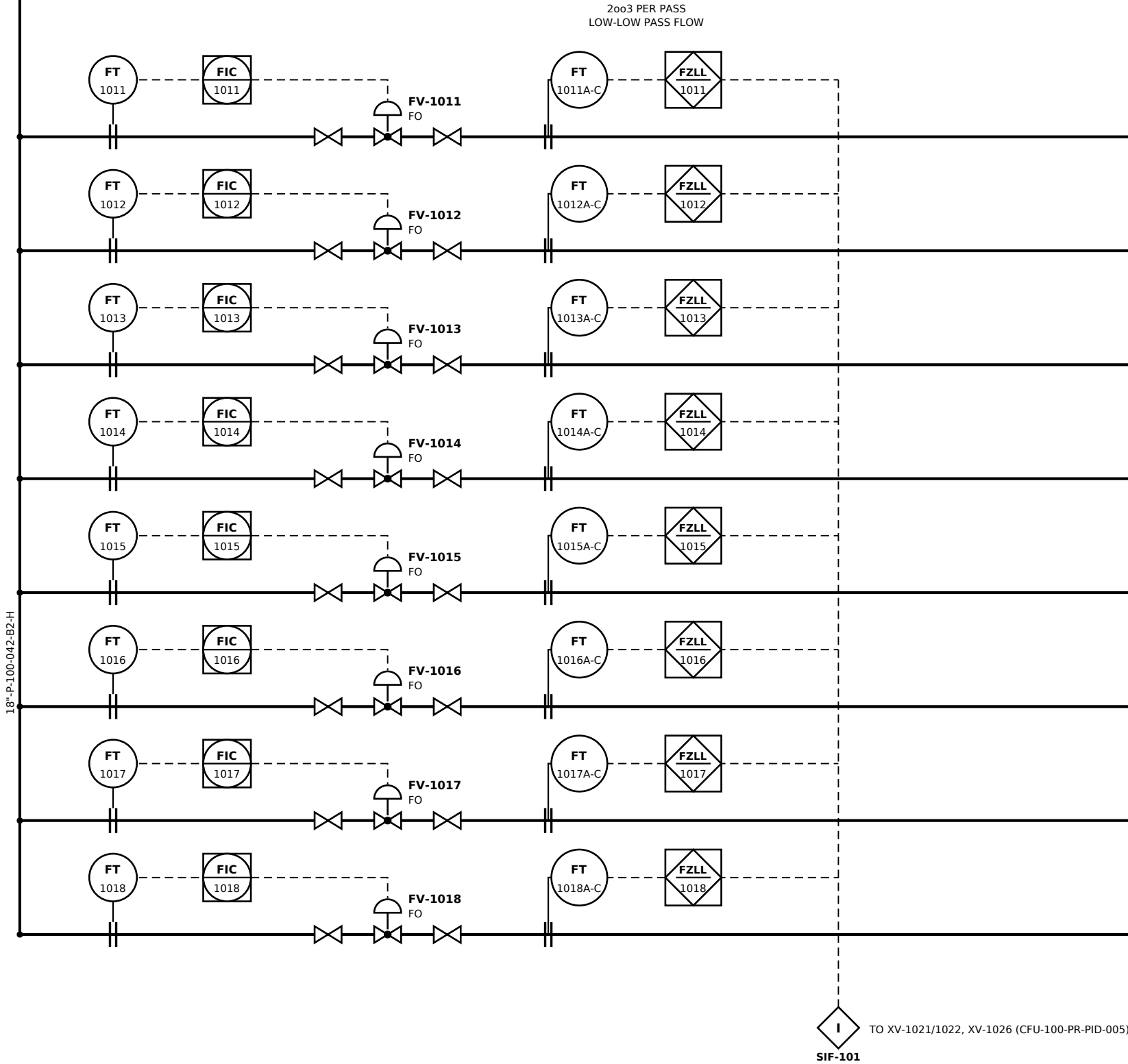
TT
1020

TIC
1020

SP CASCADE TO PIC-1021
(CFU-100-PR-PID-005); FF FROM FIC-1001

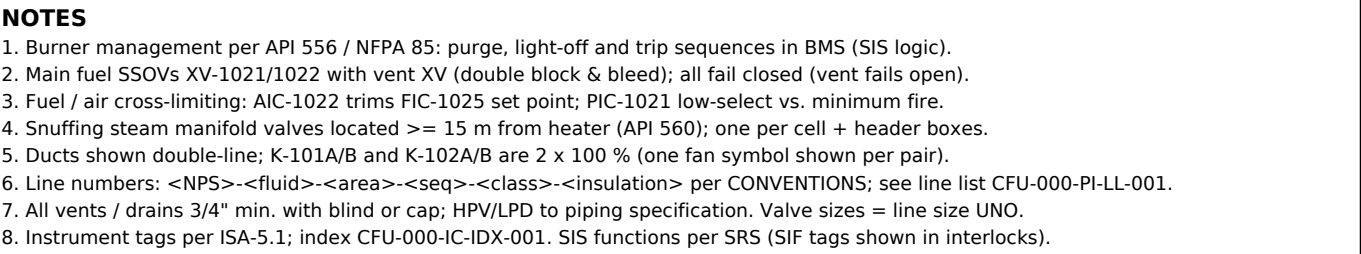
TDIC
1019

BIAS TO FIC-1011-1018
(TOTAL FLOW HELD)



- NOTES**
- Passes shown schematically (convection + radiant in series per pass); 8 identical passes.
 - Tube-skin thermocouples: 4 per pass (TI-1101..1132) - shown on heater vendor P&ID, alarmed in DCS.
 - Pass flow SIS transmitters FT-10xxA/B/C (2003 per pass) independent from BPCS FT-10xx (SIF-101).
 - Decoking (steam-air) connections and spool pieces at inlet / outlet manifolds by heater vendor.
 - Line numbers: <NPS>-<fluid>-<area>-<seq>-<class>-<insulation> per CONVENTIONS; see line list CFU-000-PI-LL-001.
 - All vents / drains 3/4" min. with blind or cap; HPV/LPD to piping specification. Valve sizes = line size UNO.
 - Instrument tags per ISA-5.1; index CFU-000-IC-IDX-001. SIS functions per SRS (SIF tags shown in interlocks).

REV	DATE	DESCRIPTION	
A	2026-10-02	Issued for review (FEED)	
GENERIC REFINERY - GULF COAST			
100 KBPSD CRUDE & VACUUM DISTILLATION UNIT			
PIPING & INSTRUMENT DIAGRAM			
ATMOSPHERIC HEATER H-101 - PROCESS COILS			
DISCIPLINE	STAGE	SCALE	SHEET
PROCESS	FEED	NTS	5 OF 17
DRAWN	CHECKED	APPROVED	DATE
CLAUDE CODE	-	-	2026-10-02
DRAWING NUMBER			REV
CFU-100-PR-PID-004			A

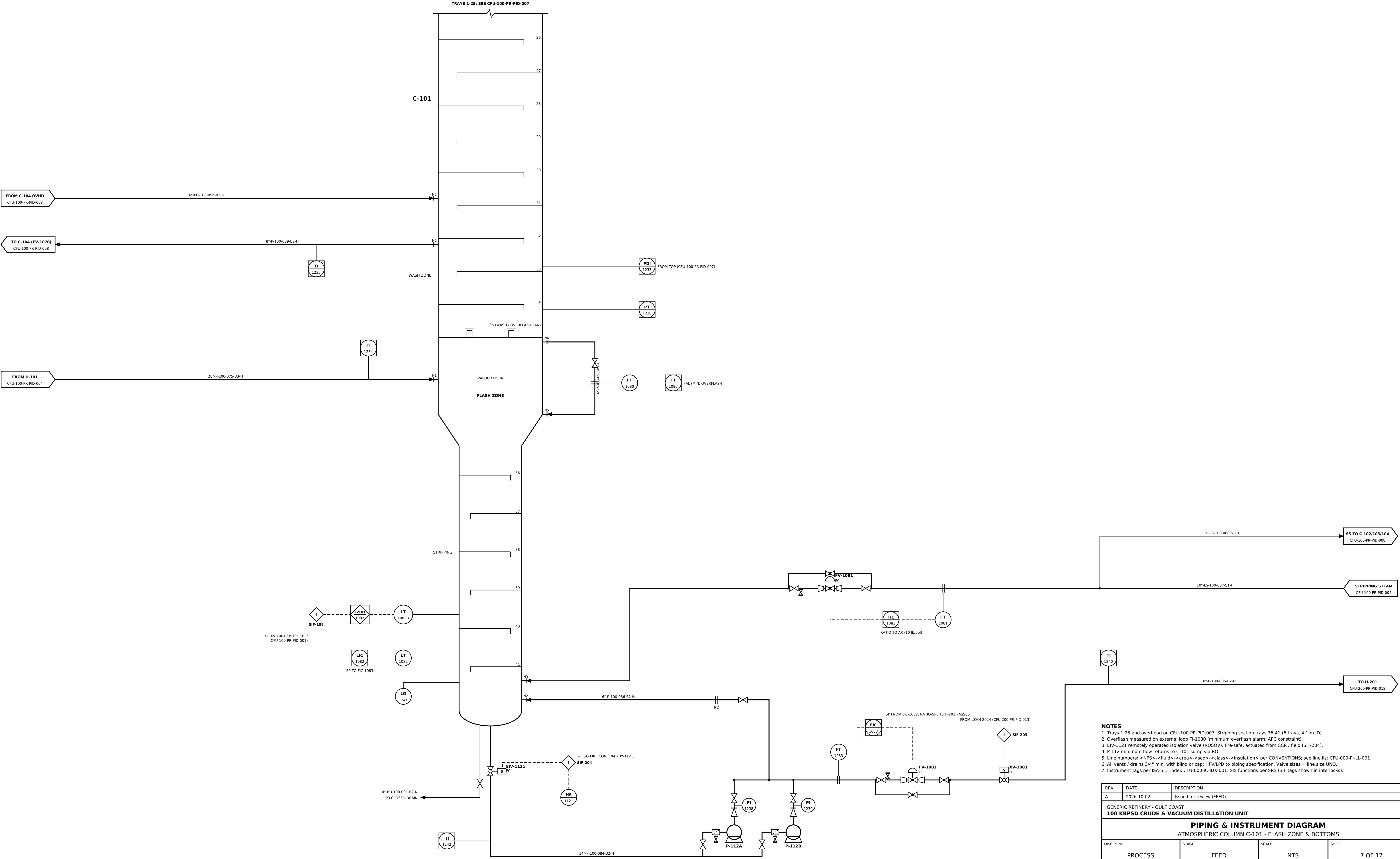


PIPING & INSTRUMENT DIAGRAM

ATMOSPHERIC HEATER H-101 - FIRING & AIR SYSTEM

C-101
Atmospheric crude fractionator
ID 6.9 m (top/main) / 3.5 m (stripping) x 37.0 m T/T
DESIGN 3.5 barg / 390 C, CA 3 mm
41 valve trays (2/4-pass), 410S; Monel-lined top 5 trays
MATL CS + 410S clad below tray 10; Monel 400 lined top head & trays 1-5

P-112A/B
Atm. residue / vacuum heater charge
RATED 398 m3/h @ 244 m, dP 17 bar
MOTOR 315 kW, API 602 Plan 538
DES. T 380 C; MATL A-8 (316)



- NOTES**
- 1. Trays 1-25 and overhead on CFU-100-PR-PID-007. Stripping section trays 36-41 (6 trays, 4.1 m ID).
 - 2. Overflash measured on external loop FI-1080 (minimum overflash alarm, APC constraint).
 - 3. EIV-1121, remotely operated isolation valve (ROSOV), fire-safe, actuated from CCR / field (SIF-204).
 - 4. P-112 minimum flow returns to C-101 sump via RO.
 - 5. Line numbers: <NPS>-<fluid>-<area>-<seq>-<class>-<insulation> per CONVENTIONS; see line list CFU-000-PI-LL-001.
 - 6. All vents / drains 3/4" min. with blind or cap; HPV/LPD to piping specification. Valve sizes = line size UNO.
 - 7. Instrument tags per ISA-5.1; index CFU-000-IC-IDX-001. SIS functions per SRS (SIF tags shown in interlocks).

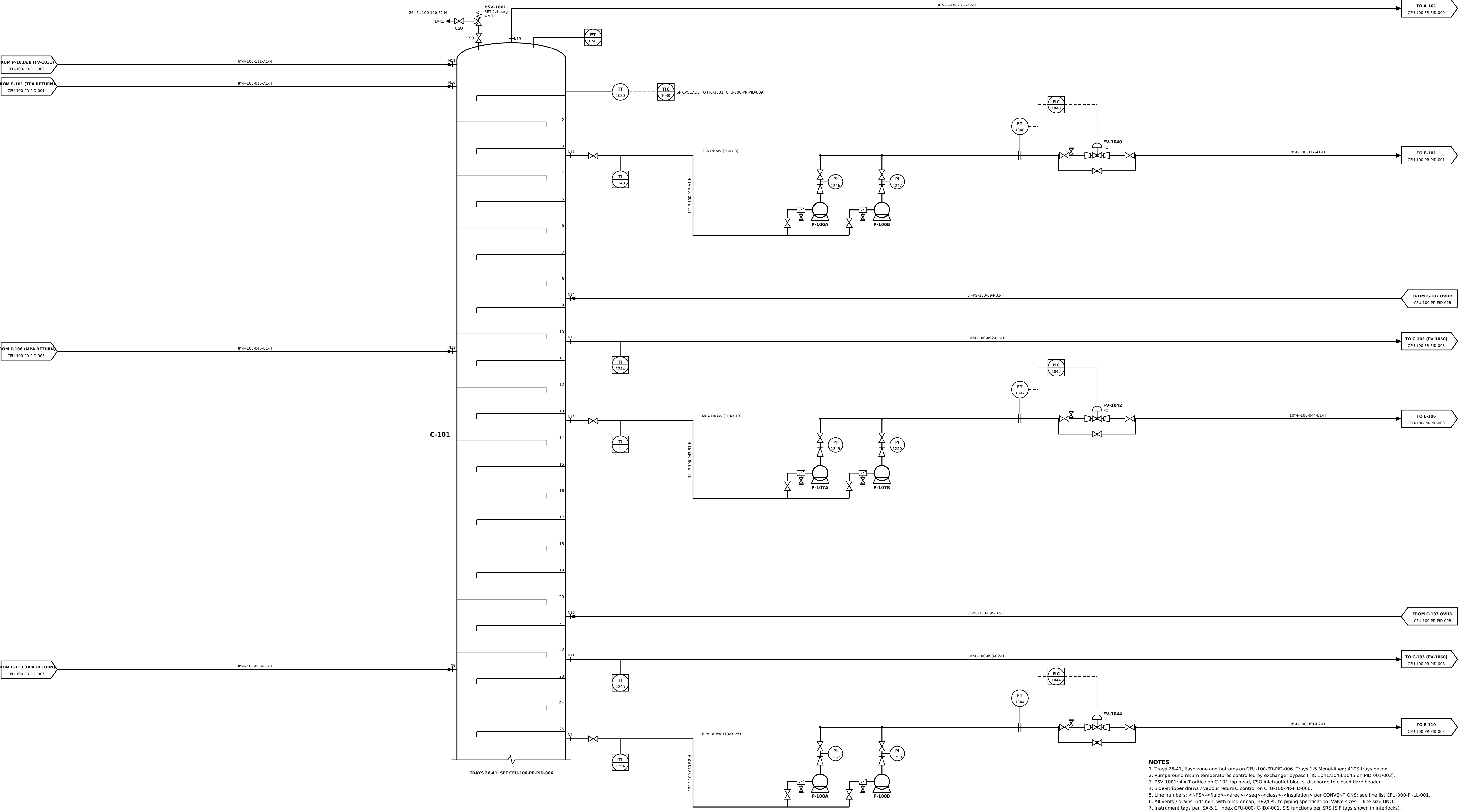
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GENERIC REFINERY - GULF COAST			
100 KBPSD CRUDE & VACUUM DISTILLATION UNIT			
PIPING & INSTRUMENT DIAGRAM			
ATMOSPHERIC COLUMN C-101 - FLASH ZONE & BOTTOMS			
DISCIPLINE	STAGE	SCALE	SHEET
PROCESS	FEED	NTS	7 OF 17
DRAWN	CHECKED	APPROVED	DATE
CLAUDE CODE	-	-	2026-10-02
DRAWING NUMBER			REV
CFU-100-PR-PID-006			A

C-101
Atmospheric crude fractionator
ID 6.9 m (top/main) / 3.5 m (stripping) x 37.0 m T/T
DESIGN 3.5 barg / 390 C; CA 3 mm
41 valve trays (2/4-pass), 4105; Monel-lined top 5 trays
MAT'L CS + 4105 clad below tray 10; Monel 400 lined top head & trays
1-5

P-106A/B
TPA pumparound
RATED 293 m³/h @ 130 m, dP 8 bar
MOTOR 1104 kW, API 682 Plus 1152
DES. T 185 C; MAT'L S-6 (CS/12Cr)

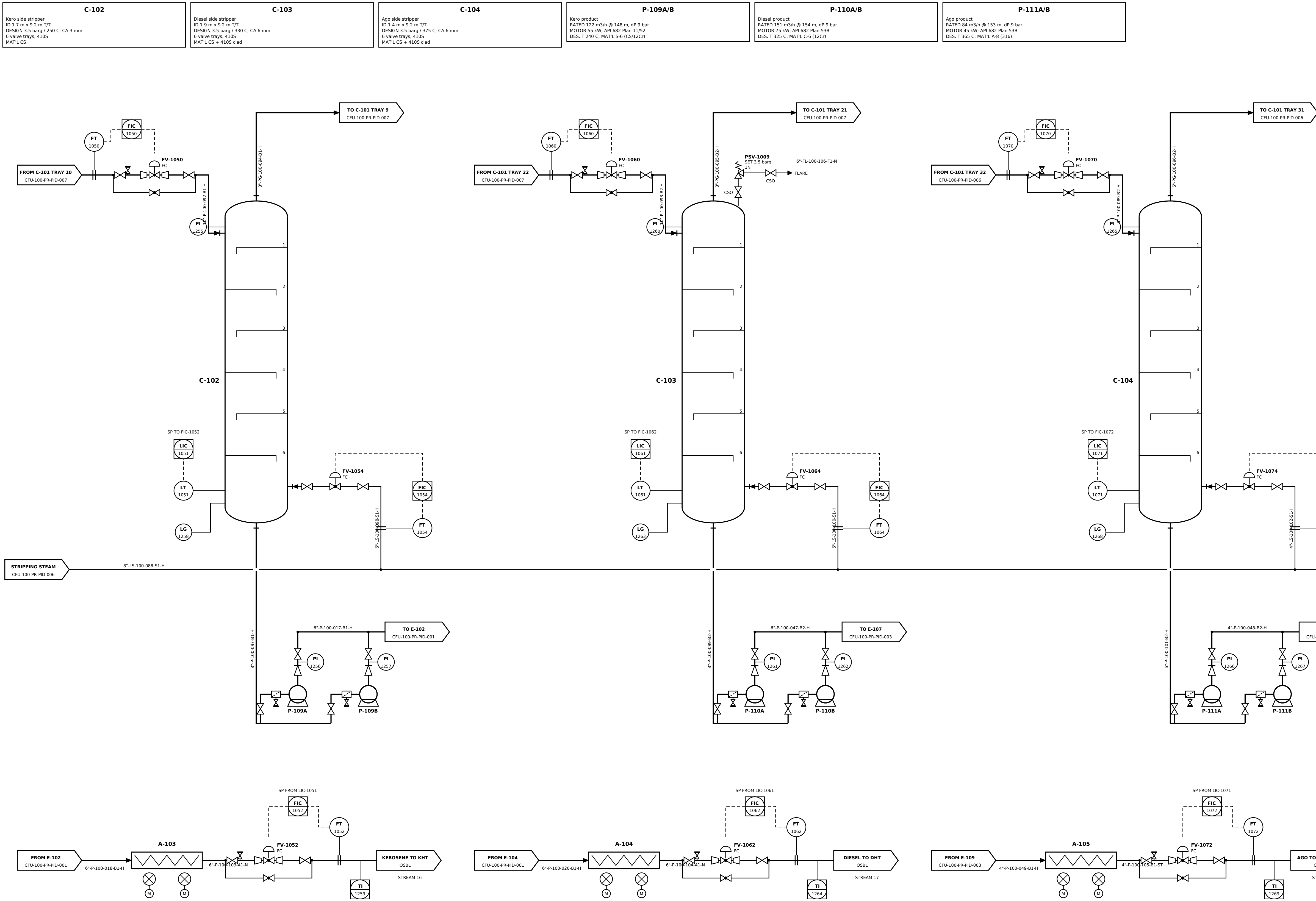
P-107A/B
MPA pumparound
RATED 387 m³/h @ 138 m, dP 8 bar
MOTOR 1364 kW, API 682 Plus 1152
DES. T 270 C; MAT'L C-6 (12Cr)

P-108A/B
BPA pumparound
RATED 318 m³/h @ 142 m, dP 8 bar
MOTOR 1324 kW, API 682 Plus 538
DES. T 345 C; MAT'L A-8 (316)



- NOTES**
- Trays 26-41, flash zone and bottoms on CFU-100-PR-PID-006. Trays 1-5 Monel-lined; 4105 trays below.
 - Pumparound return temperatures controlled by exchanger bypass (TIC-1041/1043/1045 on PID-001/003).
 - PSV-1001: 4 x T orifice on C-101 top head. CSO inlet/outlet blocks: discharge to closed flare header.
 - Side-stripper draws / vapour returns: control on CFU-100-PR-PID-008.
 - Line numbers: <NPS>-<fluid>-<area>-<seq>-<class>-<insulation> per CONVENTIONS; see line list CFU-000-PI-LL-001.
 - All vents / drains 3/4" min. with blind or cap; HPV/LPD to piping specification. Valve sizes = line size UNO.
 - Instrument tags per ISA-5.1; index CFU-000-IC-IDX-001. SIS functions per SRS (SIF tags shown in interlocks).

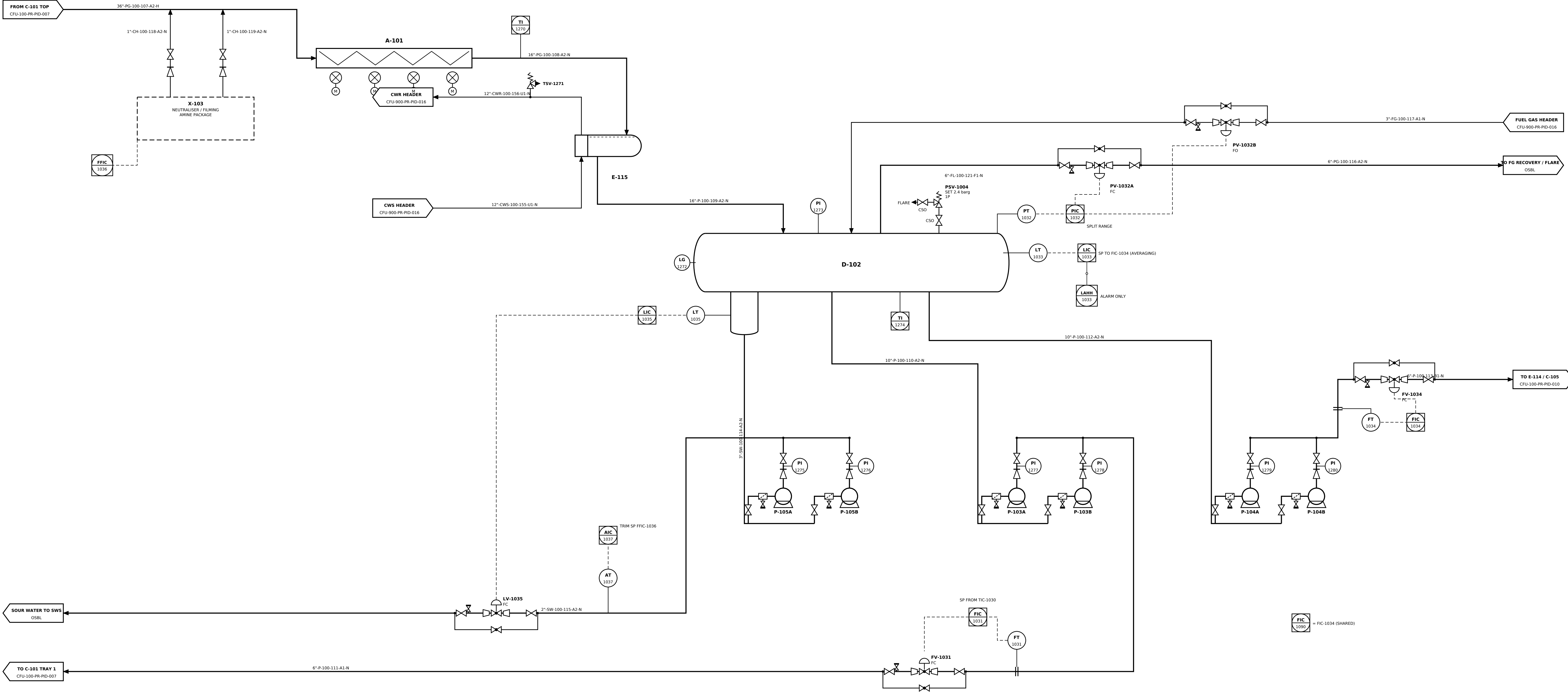
REV	DATE	DESCRIPTION	
A	2026-10-02	Issued for review (FEED)	
GENERIC REFINERY - GULF COAST			
100 KBPSD CRUDE & VACUUM DISTILLATION UNIT			
PIPING & INSTRUMENT DIAGRAM			
ATMOSPHERIC COLUMN C-101 - UPPER SECTION & PUMPAROUNDS			
DISCIPLINE	STAGE	SCALE	SHEET
PROCESS	FEED	NTS	8 OF 17
DRAWN	CHECKED	APPROVED	DATE
CLAUDE CODE	-	-	2026-10-02
DRAWING NUMBER			REV
CFU-100-PR-PID-007			A



- NOTES**
- Side-stripper vapours return to C-101 one tray above draw; strippers protected by PSV-1009 (fire case).
 - Stripping steam ratio to product rate (6 lb/bbl) via FIC-1054/1064/1074.
 - Product rundown temperatures: kerosene 45 C, diesel 55 C, AGO 60 C (air coolers with VFD on one fan).
 - Line numbers: <NPS>-<fluid>-<area>-<seq>-<class>-<insulation> per CONVENTIONS; see line list CFU-000-PI-LL-001.
 - All vents / drains 3/4" min. with blind or cap; HPV/LPD to piping specification. Valve sizes = line size UNO.
 - Instrument tags per ISA-5.1; index CFU-000-IC-IDX-001. SIS functions per SRS (SIF tags shown in interlocks).

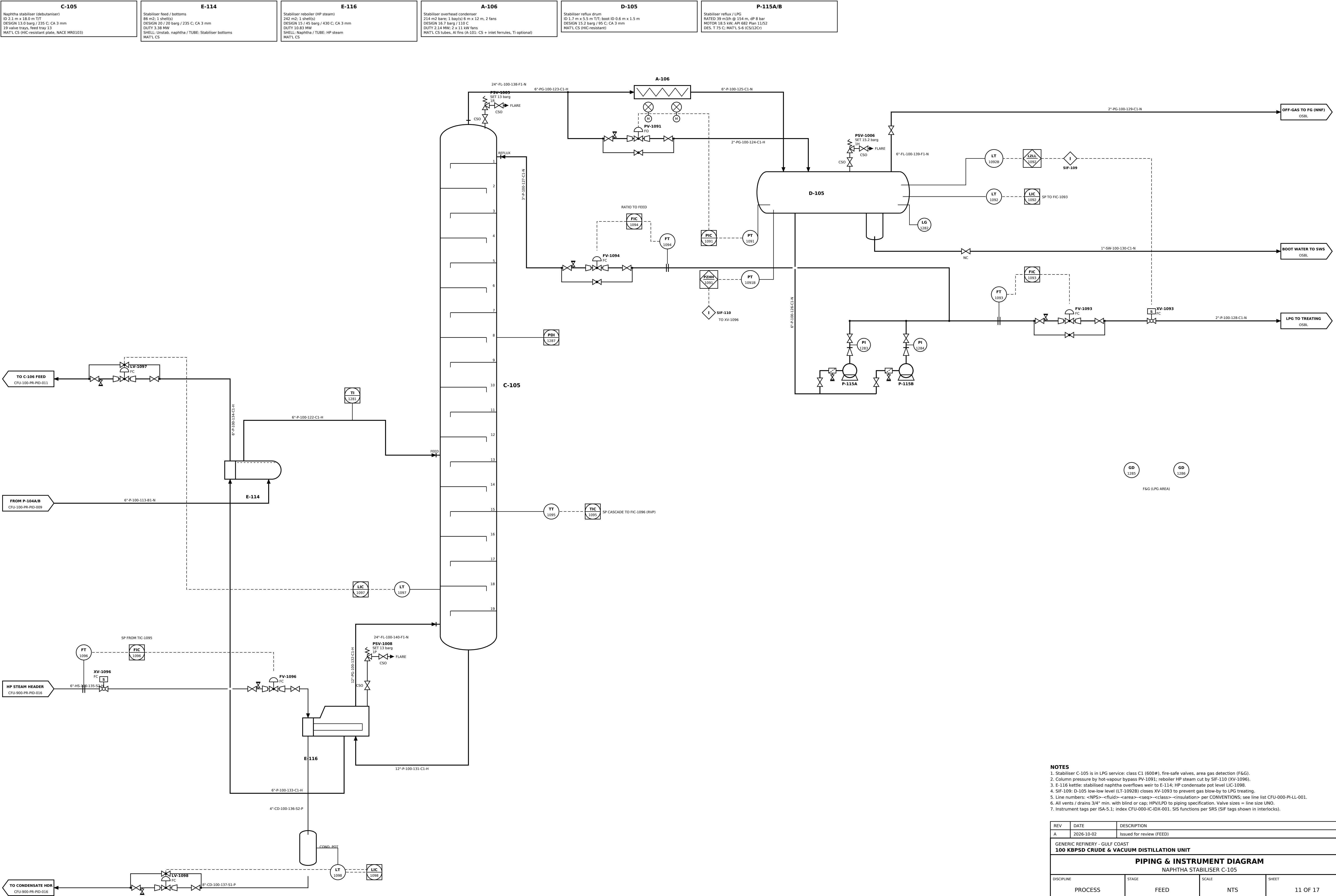
REV	DATE	DESCRIPTION	
A	2026-10-02	Issued for review (FEED)	
GENERIC REFINERY - GULF COAST 100 KBPSD CRUDE & VACUUM DISTILLATION UNIT			
PIPING & INSTRUMENT DIAGRAM SIDE STRIPPERS C-102/103/104 & PRODUCT RUNDOWN			
DISCIPLINE	STAGE	SCALE	SHEET
PROCESS	FEED	NTS	9 OF 17
DRAWN	CHECKED	APPROVED	DATE
CLAUDE CODE	-	-	2026-10-02
DRAWING NUMBER			REV
CFU-100-PR-PID-008			A

A-101 Atm. overhead condenser 1667 m2 bare; 3 bay(s) 6 m x 12 m, 6 fans DESIGN 9.7 barg / 165 C DUTY 33.75 MW; 6 x 55 kW fans MAT'L CS tubes, Al fins (A-101: CS + inlet ferrules, Ti optional)	E-115 Atm. overhead trim condenser 740 m2; 2 shell(s) DESIGN 3.5 / 7 barg / 90 C; CA 3 mm DUTY 5.96 MW SHELL: OH vapour/condensate / TUBE: Cooling water MAT'L CS shell / Ti tubes	D-102 Atm. overhead reflux drum (3-phase) ID 3.4 m x 12.0 m T/T; boot ID 1.5 m x 3.0 m DESIGN 3.5 barg / 160 C; CA 6 mm MAT'L CS (HIC-resistant), 6 mm CA	P-103A/B Atm. reflux RATED 159 m3/h @ 102 m, dP 7 bar MOTOR: 55 kW; API 682 Plan 11/52 DES. T 75 C; MAT'L S-6 (CS/12Cr)	P-104A/B Unstabilised naphtha RATED 180 m3/h @ 232 m, dP 16 bar MOTOR: 160 kW; API 682 Plan 11/52 DES. T 75 C; MAT'L S-6 (CS/12Cr)	P-105A/B Overhead sour water RATED 14 m3/h @ 51 m, dP 2 bar MOTOR: 5.5 kW; API 682 Plan 11/52 DES. T 75 C; MAT'L S-6 (CS/12Cr)	X-103 Neutraliser / filming amine package (atm OH) Tank 2 m3 + 2 x 100 % metering pumps MOTOR: 0.75 kW MAT'L SS316
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- NOTES**
- Neutraliser and filming amine injected upstream of A-101; pH control AIC-1037 trims FFIC-1036.
 - PIC-1032 split range: 0-50 % PV-1032B (FG make-up) closing, 50-100 % PV-1032A (off-gas) opening.
 - D-102 high-high level is a DCS alarm only (LAHH-1033); TSV on E-115 CW side, set per TSV-tyt.
 - Line numbers: <NPS>-<fluid>-<area>-<seq>-<class>-<insulation> per CONVENTIONS; see line list CFU-000-PI-LL-001.
 - All vents / drains 3/4" min. with blind or cap; HPV/LPD to piping specification. Valve sizes = line size UNO.
 - Instrument tags per ISA-5.1; index CFU-000-IC-IDX-001. SIS functions per SRS (SIF tags shown in interlocks).

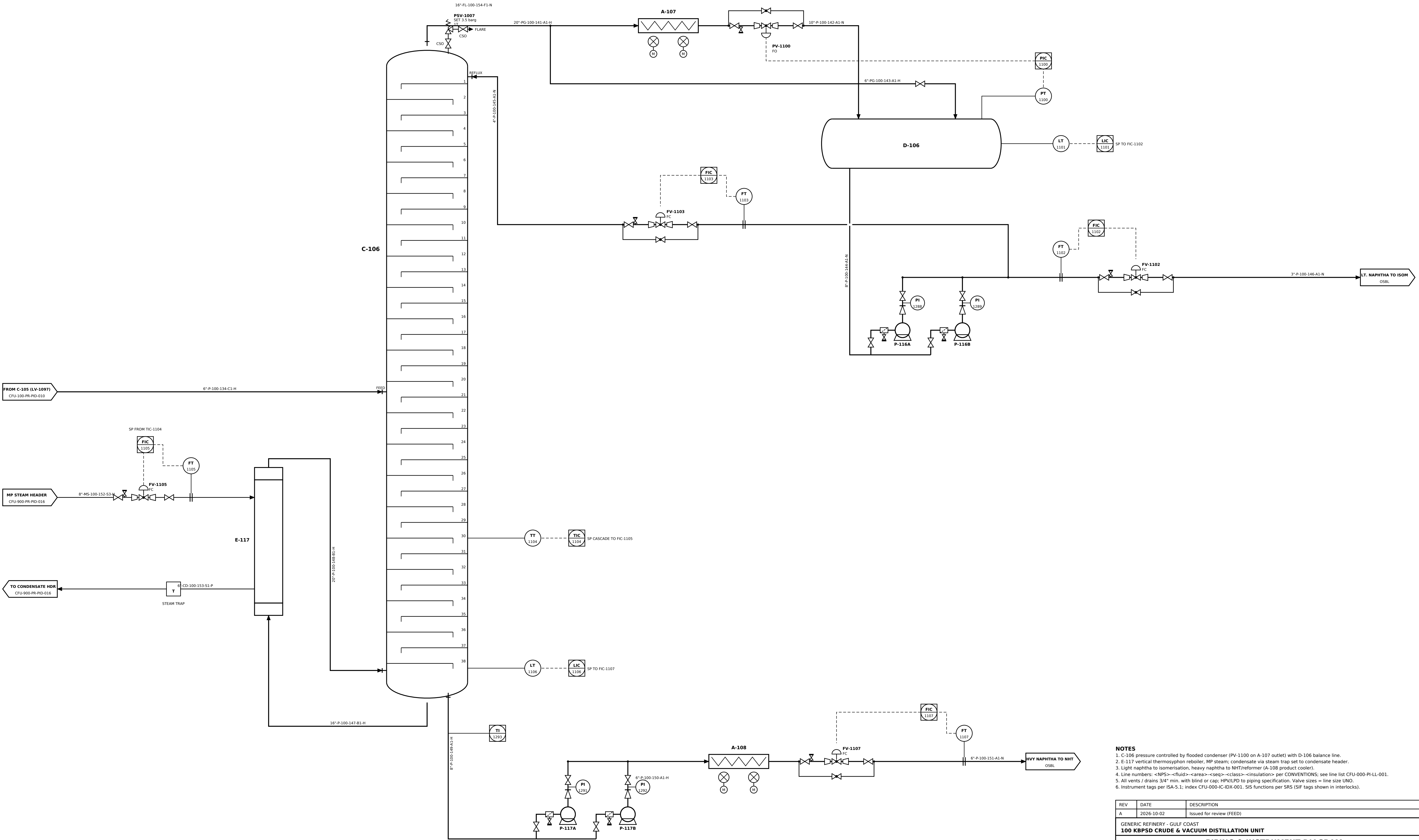
REV	DATE	DESCRIPTION		
A	2026-10-02	Issued for review (FEED)		
GENERIC REFINERY - GULF COAST 100 KBPSD CRUDE & VACUUM DISTILLATION UNIT				
PIPING & INSTRUMENT DIAGRAM C-101 OVERHEAD SYSTEM & REFLUX DRUM D-102				
DISCIPLINE		STAGE	SCALE	SHEET
PROCESS		FEED	NTS	10 OF 17
DRAWN	CHECKED	APPROVED		DATE
CLAUDE CODE	-	-		2026-10-02
DRAWING NUMBER				REV
CFU-100-PR-PID-009				A



- NOTES**
- 1. Stabiliser C-105 is in LPG service: class C1 (600#), fire-safe valves, area gas detection (F&G).
 - 2. Column pressure by hot-vapour bypass PV-1091; reboiler HP steam cut by SIF-110 (XV-1096).
 - 3. E-116 kettle: stabilised naphtha overflows weir to E-114; HP condensate pot level LIC-1098.
 - 4. SIF-109: D-105 low-low level (LT-1092B) closes XV-1093 to prevent gas blow-by to LPG treating.
 - 5. Line numbers: <NPS>-<fluid>-<area>-<seq>-<class>-<insulation> per CONVENTIONS; see line list CFU-000-PI-LL-001.
 - 6. All vents / drains 3/4" min. with blind or cap; HPV/LPD to piping specification. Valve sizes = line size UNO.
 - 7. Instrument tags per ISA-5.1; index CFU-000-IC-IDX-001. SIS functions per SRS (SIF tags shown in interlocks).

REV	DATE	DESCRIPTION	
A	2026-10-02	Issued for review (FEED)	
GENERIC REFINERY - GULF COAST			
100 KBPSD CRUDE & VACUUM DISTILLATION UNIT			
PIPING & INSTRUMENT DIAGRAM			
NAPHTHA STABILISER C-105			
DISCIPLINE	STAGE	SCALE	SHEET
PROCESS	FEED	NTS	11 OF 17
DRAWN	CHECKED	APPROVED	DATE
CLAUDE CODE	-	-	2026-10-02
DRAWING NUMBER			REV
CFU-100-PR-PID-010			A

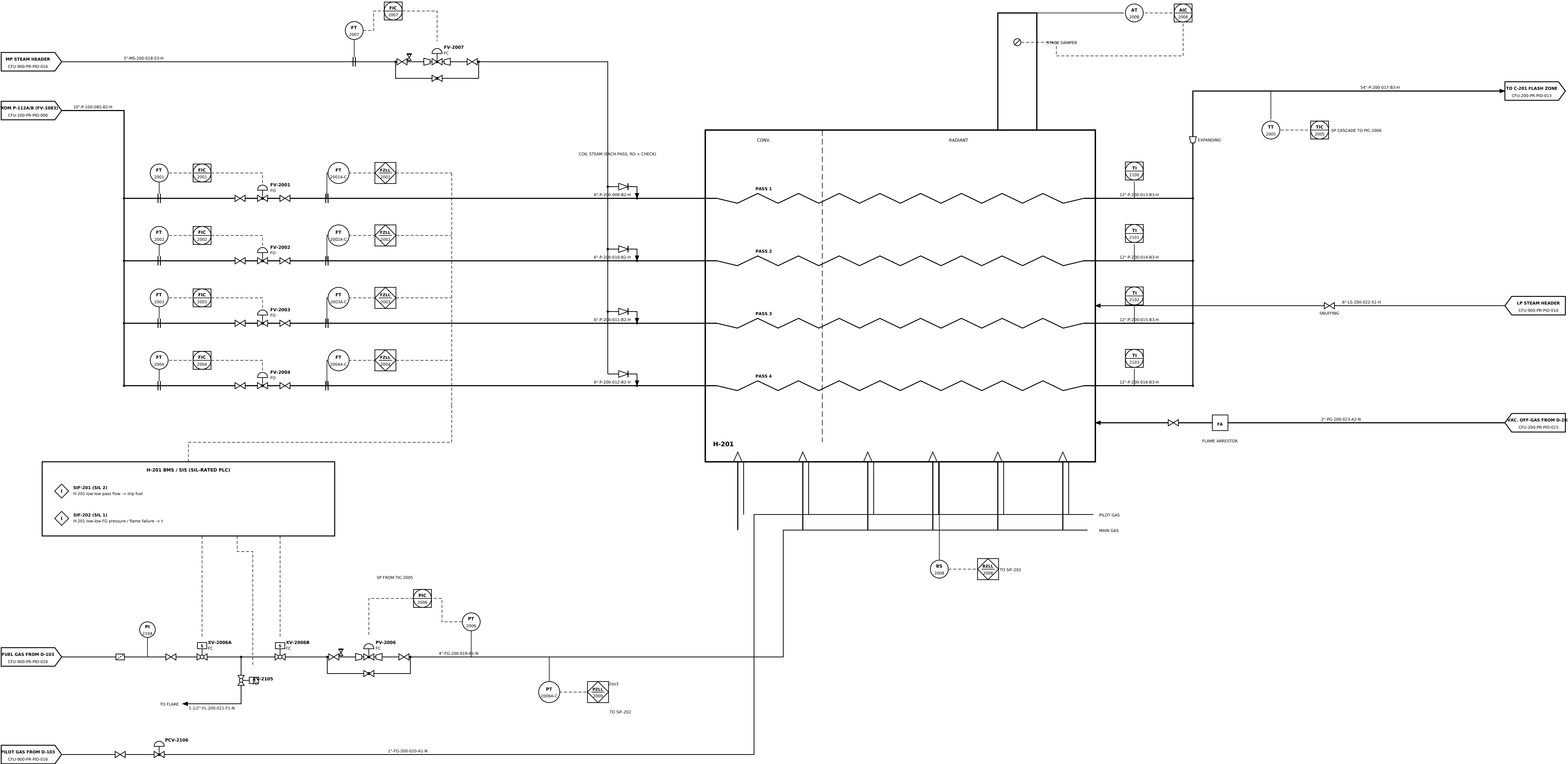
C-106 Naphtha splitter ID 3.1 m x 29.5 m TTT DESIGN 3.5 barg / 185 C; CA 3 mm 38 valve trays, feed tray 21 MAT'L CS	E-117 Splitter reboiler (MP steam) 225 m2; 1 shell(s) DESIGN 5 / 13 barg / 280 C; CA 3 mm DUTY 6.83 MW SHELL: Naphtha / TUBE: MP steam MAT'L CS	A-107 Splitter overhead condenser BB1 m2 bare; 2 bay(s) 6 m x 12 m, 4 fans DESIGN 9.7 barg / 135 C DUTY 9.63 MW; 4 x 30 kW fans MAT'L CS tubes, Al fins (A-101; CS + inlet ferrules, Ti optional)	D-106 Splitter reflux drum ID 2.7 m x 8.5 m TTT DESIGN 3.5 barg / 120 C; CA 3 mm MAT'L CS	P-116A/B Splitter reflux / LN RATED 146 m3/h @ 111 m, dP 7 bar MOTOR 55 kW, API 682 Flan 11/52 DES. T 80 C; MAT'L S-6 (CS/12Cr)	P-117A/B Heavy naphtha product RATED 142 m3/h @ 98 m, dP 6 bar MOTOR 45 kW, API 682 Flan 11/52 DES. T 185 C; MAT'L S-6 (CS/12Cr)	A-108 Heavy naphtha product cooler 401 m2 bare; 1 bay(s) 6 m x 12 m, 2 fans DESIGN 9.7 barg / 185 C DUTY 5.81 MW; 2 x 30 kW fans MAT'L CS tubes, Al fins (A-101; CS + inlet ferrules, Ti optional)
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- NOTES**
- C-106 pressure controlled by flooded condenser (PV-1100 on A-107 outlet) with D-106 balance line.
 - E-117 vertical thermosiphon reboiler, MP steam; condensate via steam trap set to condensate header.
 - Light naphtha to isomerisation, heavy naphtha to NHT/reformer (A-108 product cooler).
 - Line numbers: <NPS>-<fluid>-<area>-<seq>-<class>-<insulation> per CONVENTIONS; see line list CFU-000-PI-LL-001.
 - All vents / drains 3/4" min. with blind or cap; HPV/LPD to piping specification. Valve sizes = line size UNO.
 - Instrument tags per ISA-5.1; index CFU-000-IC-IDX-001. SIS functions per SRS (SIF tags shown in interlocks).

REV	DATE	DESCRIPTION	
A	2026-10-02	Issued for review (FEED)	
GENERIC REFINERY - GULF COAST 100 KBPSD CRUDE & VACUUM DISTILLATION UNIT			
PIPING & INSTRUMENT DIAGRAM NAPHTHA SPLITTER C-106			
DISCIPLINE		STAGE	SCALE
PROCESS		FEED	NTS
			12 OF 17
DRAWN		CHECKED	DATE
CLAUDE CODE		-	2026-10-02
DRAWING NUMBER			REV
CFU-100-PR-PID-011			A

H-201
Vacuum heater
14.7 MW abs. / 17.5 MW fired; 1-cell cabin, 4 passes, 44 rad. tubes 6" x 18.3 m
DESIGN 19.8 barg / 580 C; CA 3 mm
DUTY 14.69 MW
MAT'L Tubes A335 P9 (9Cr-1Mo); convection P5 / CS studded



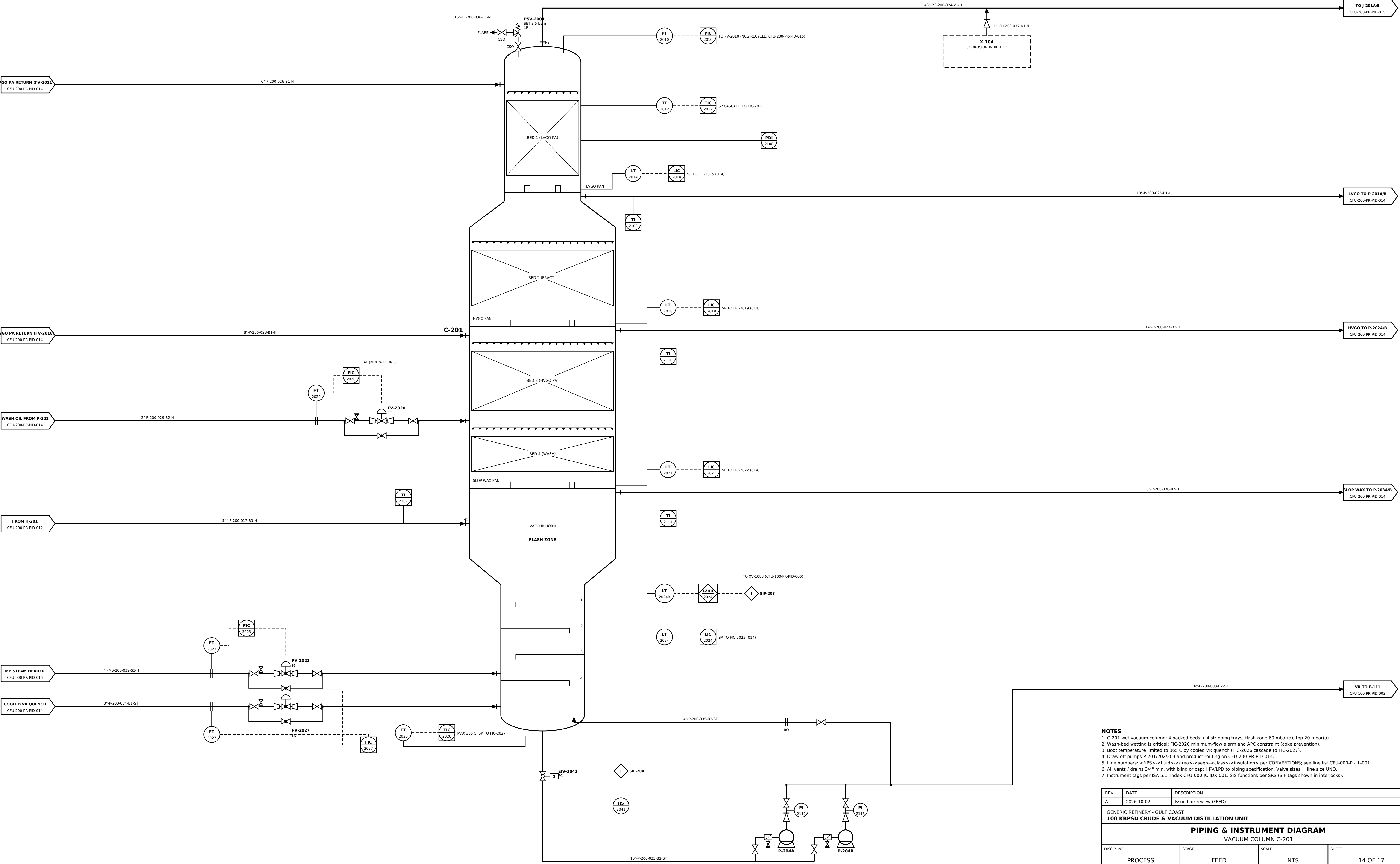
- NOTES**
- Passes shown schematically (convection + radiant in series); 4 identical passes, coil steam per pass.
 - Pass flow SIS transmitters FT-200xA/B/C (2003) independent of BPCS FT-200x (SIF-201).
 - Vacuum off-gas burned in dedicated H-201 burners via flame arrestor (from D-202 / PV-2031).
 - Natural draft heater: O2 trim AIC-2008 on stack damper.
 - Line numbers: <NPS>-<fluid>-<area>-<seq>-<class>-<insulation> per CONVENTIONS; see line list CFU-000-PI-LL-001.
 - All vents / drains 3/4" min. with blind or cap; HPV/LPD to piping specification. Valve sizes = line size UNO.
 - Instrument tags per ISA-5.1; index CFU-000-IC-IDX-001. SIS functions per SRS (SIF tags shown in interlocks).

REV	DATE	DESCRIPTION	
A	2026-10-02	Issued for review (FEED)	
GENERIC REFINERY - GULF COAST			
100 KBPSD CRUDE & VACUUM DISTILLATION UNIT			
PIPING & INSTRUMENT DIAGRAM			
VACUUM HEATER H-201			
DISCIPLINE		STAGE	SCALE
PROCESS		FEED	NTS
		SHEET	
		13 OF 17	
DRAWN	CHECKED	APPROVED	DATE
CLAUDE CODE	-	-	2026-10-02
DRAWING NUMBER			REV
CFU-200-PR-PID-012			A

C-201
Vacuum column (wet, packed)
ID 4.4 m (top) / 6.6 m (main) / 3.0 m (boot) x 36.0 m T/T
DESIGN PV: 3.5 barg / 435 C, CA 6 mm
4 packed beds (structured 250V/grid), 4 stripping trays, vapour horn
MATL: CS + 410S clad (317L clad flash/wash zone)

P-204A/B
Vacuum residue (incl. quench)
RATED 196 m3/h @ 187 m, dP 14 bar
MOTOR: 132 kW, API 682, Plan 538
DES. T 395 C, MATL: A-8 (316)

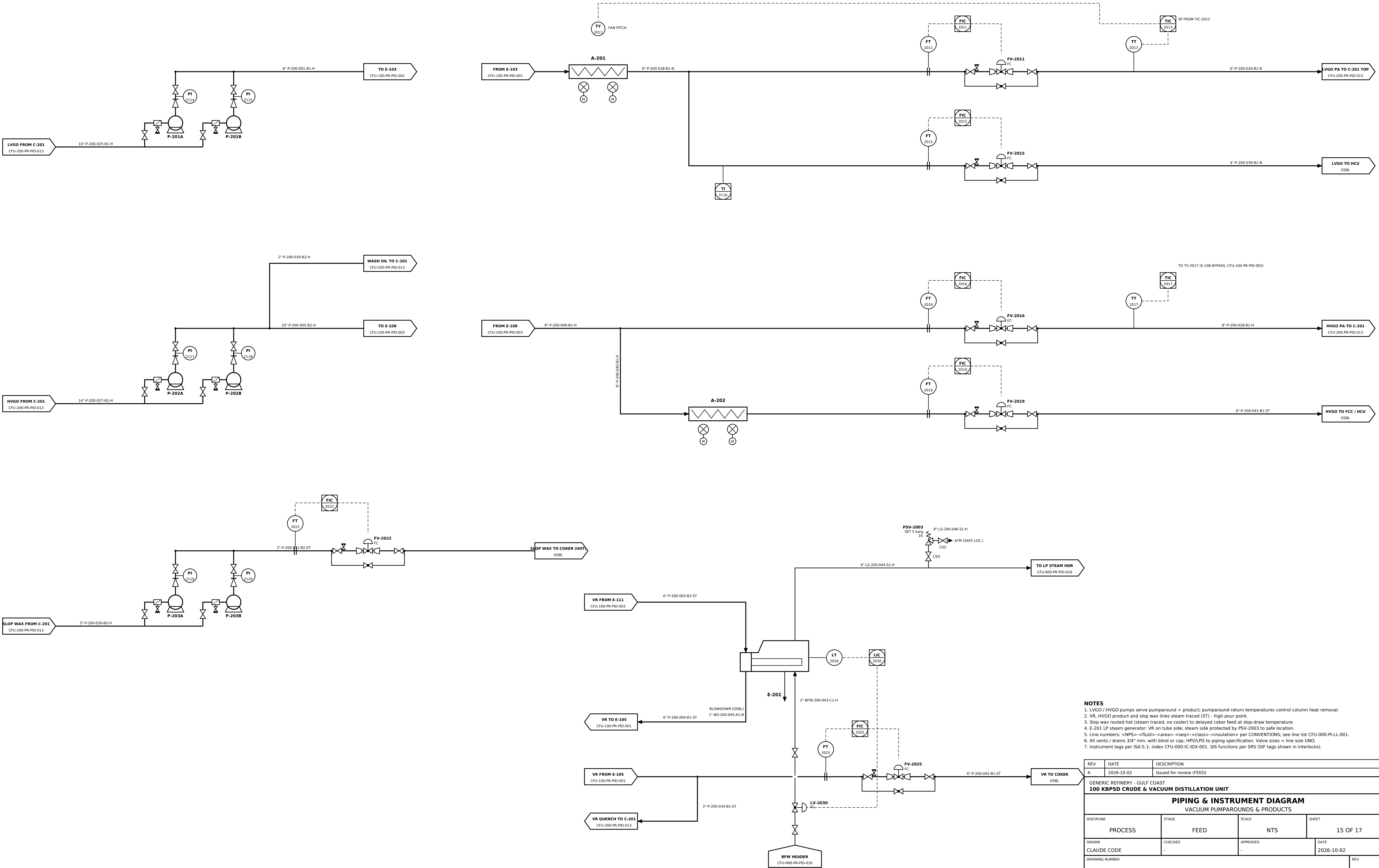
X-104
Corrosion inhibitor package (VDU OH)
Tank 2 m3 + 2 x 100 % metering pumps
MOTOR: 0.75 kW
MATL: SS316

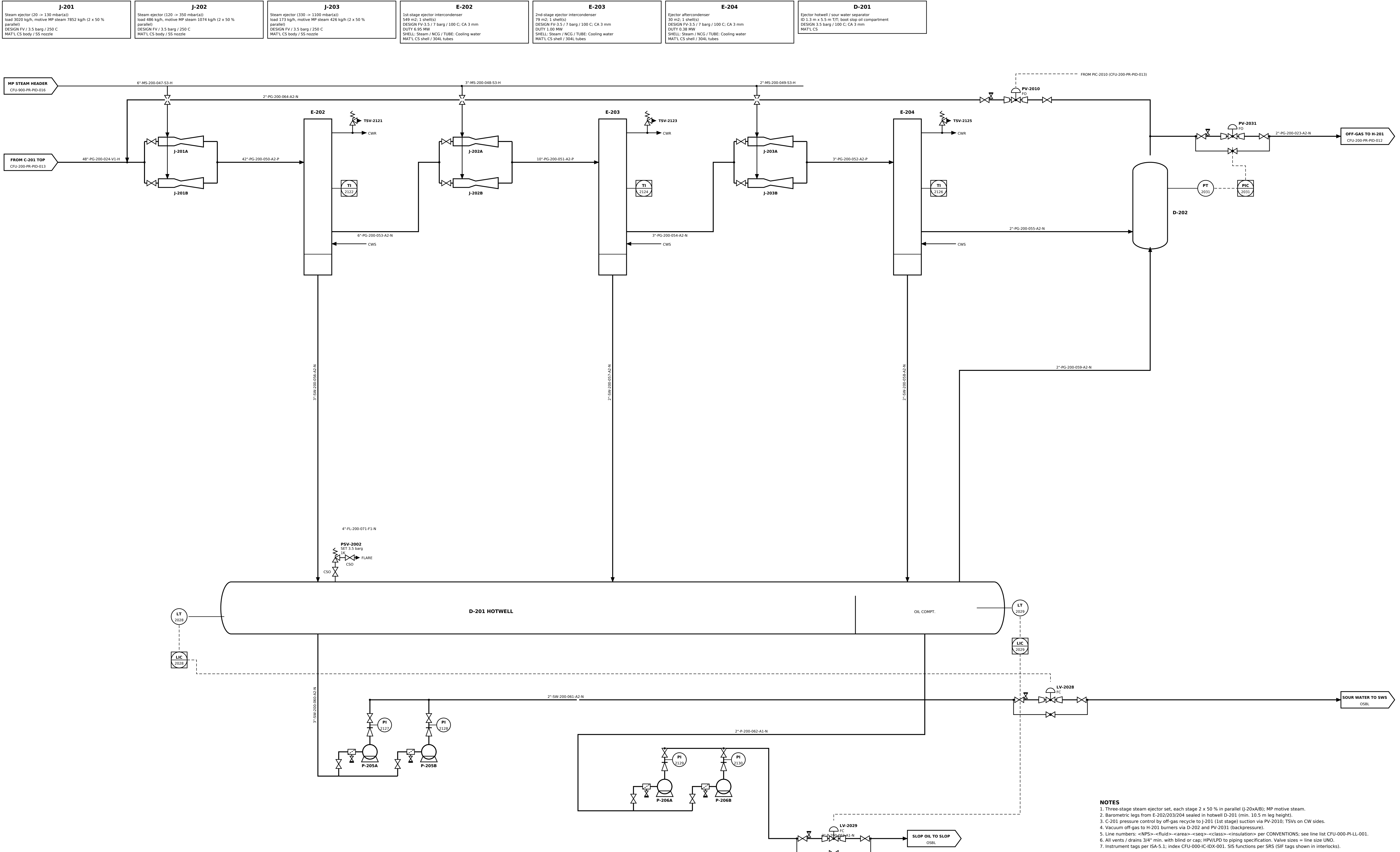


- NOTES**
- C-201 wet vacuum column: 4 packed beds + 4 stripping trays; flash zone 60 mbar(a), top 20 mbar(a).
 - Wash-bed wetting is critical: FIC-2020 minimum-flow alarm and APC constraint (coke prevention).
 - Boot temperature limited to 365 C by cooled VR quench (TIC-2026 cascade to FIC-2027).
 - Draw-off pumps P-201/202/203 and product routing on CFU-200-PR-PID-014.
 - Line numbers: <NPS>-<fluid>-<area>-<seq>-<class>-<insulation> per CONVENTIONS; see line list CFU-000-PI-LL-001.
 - All vents / drains 3/4" min. with blind or cap; HPV/LPD to piping specification. Valve sizes = line size UNO.
 - Instrument tags per ISA-5.1; index CFU-000-IC-IDX-001. SIS functions per SRS (SIF tags shown in interlocks).

REV	DATE	DESCRIPTION
A	2026-10-02	Issued for review (FEED)
GENERIC REFINERY - GULF COAST		
100 KBPSD CRUDE & VACUUM DISTILLATION UNIT		
PIPING & INSTRUMENT DIAGRAM		
VACUUM COLUMN C-201		
DISCIPLINE	STAGE	SCALE
PROCESS	FEED	NTS
DRAWN		CHECKED
CLAUDE CODE		APPROVED
DRAWING NUMBER		DATE
CFU-200-PR-PID-013		2026-10-02
REV		14 OF 17
A		

P-201A/B LVGO pumparound / product RATED 198 m3/h @ 138 m, dP 10 bar MOTOR 110 kW, API 682 Plan 1122 DES. T 255 C; MAT'L S-6 (CS/12Cr)	P-202A/B HVGO pumparound / product RATED 391 m3/h @ 144 m, dP 10 bar MOTOR 200 kW, API 682 Plan 538 DES. T 345 C; MAT'L A-8 (316)	P-203A/B Slop wax RATED 12 m3/h @ 114 m, dP 8 bar MOTOR 5.5 kW, API 682 Plan 538 DES. T 410 C; MAT'L A-8 (316)	A-201 Lvgo cooler 397 m2 bare; 1 bay(s) 6 m x 12 m, 2 fans DESIGN 9.7 barg / 125 C DUTY 3.10 MW; 2 x 15 kW fans MAT'L CS tubes, Al fins (A-101; CS + inlet ferrules, Ti optional)	A-202 Hvgo Product cooler 361 m2 bare; 1 bay(s) 6 m x 12 m, 2 fans DESIGN 9.7 barg / 125 C DUTY 6.64 MW; 2 x 37 kW fans SHELL: BFW / LP steam / TUBE: Vacuum residue MAT'L CS tubes, Al fins (A-101; CS + inlet ferrules, Ti optional)	E-201 VR / LP steam generator (kettle) 84 m2; 1 shell(s) DESIGN 6 / 20 barg / 310 C; CA 3 mm DUTY 1.78 MW SHELL: BFW / LP steam / TUBE: Vacuum residue MAT'L CS shell / Scr tubes
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COOLING WATER CONNECTIONS (HEADERS ON CFU-900-PR-PID-016):
E-202: CWS 12" CWS-200-065-U1-N / CWR 12" CWR-200-066-U1-N
E-203: CWS 6" CWS-200-067-U1-N / CWR 6" CWR-200-068-U1-N
E-204: CWS 3" CWS-200-069-U1-N / CWR 3" CWR-200-070-U1-N

- NOTES**
- Three-stage steam ejector set, each stage 2 x 50 % in parallel (J-20xA/B); MP motive steam.
 - Barometric legs from E-202/203/204 sealed in hotwell D-201 (min. 10.5 m leg height).
 - C-201 pressure control by off-gas recycle to J-201 (1st stage) suction via PV-2010; TSVs on CW sides.
 - Vacuum off-gas to H-201 burners via D-202 and PV-2031 (backpressure).
 - Line numbers: <NPS>-<fluid>-<area>-<seq>-<class>-<insulation> per CONVENTIONS; see line list CFU-000-PI-LL-001.
 - All vents / drains 3/4" min. with blind or cap; HPV/LPD to piping specification. Valve sizes = line size UNO.
 - Instrument tags per ISA-5.1; index CFU-000-IC-IDX-001. SIS functions per SRS (SIF tags shown in interlocks).

REV	DATE	DESCRIPTION	
A	2026-10-02	Issued for review (FEED)	
GENERIC REFINERY - GULF COAST			
100 KBPSD CRUDE & VACUUM DISTILLATION UNIT			
PIPING & INSTRUMENT DIAGRAM			
VACUUM EJECTOR SYSTEM & HOTWELL D-201			
DISCIPLINE	STAGE	SCALE	SHEET
PROCESS	FEED	NTS	16 OF 17
DRAWN	CHECKED	APPROVED	DATE
CLAUDE CODE	-	-	2026-10-02
DRAWING NUMBER			REV
CFU-200-PR-PID-015			A

D-103

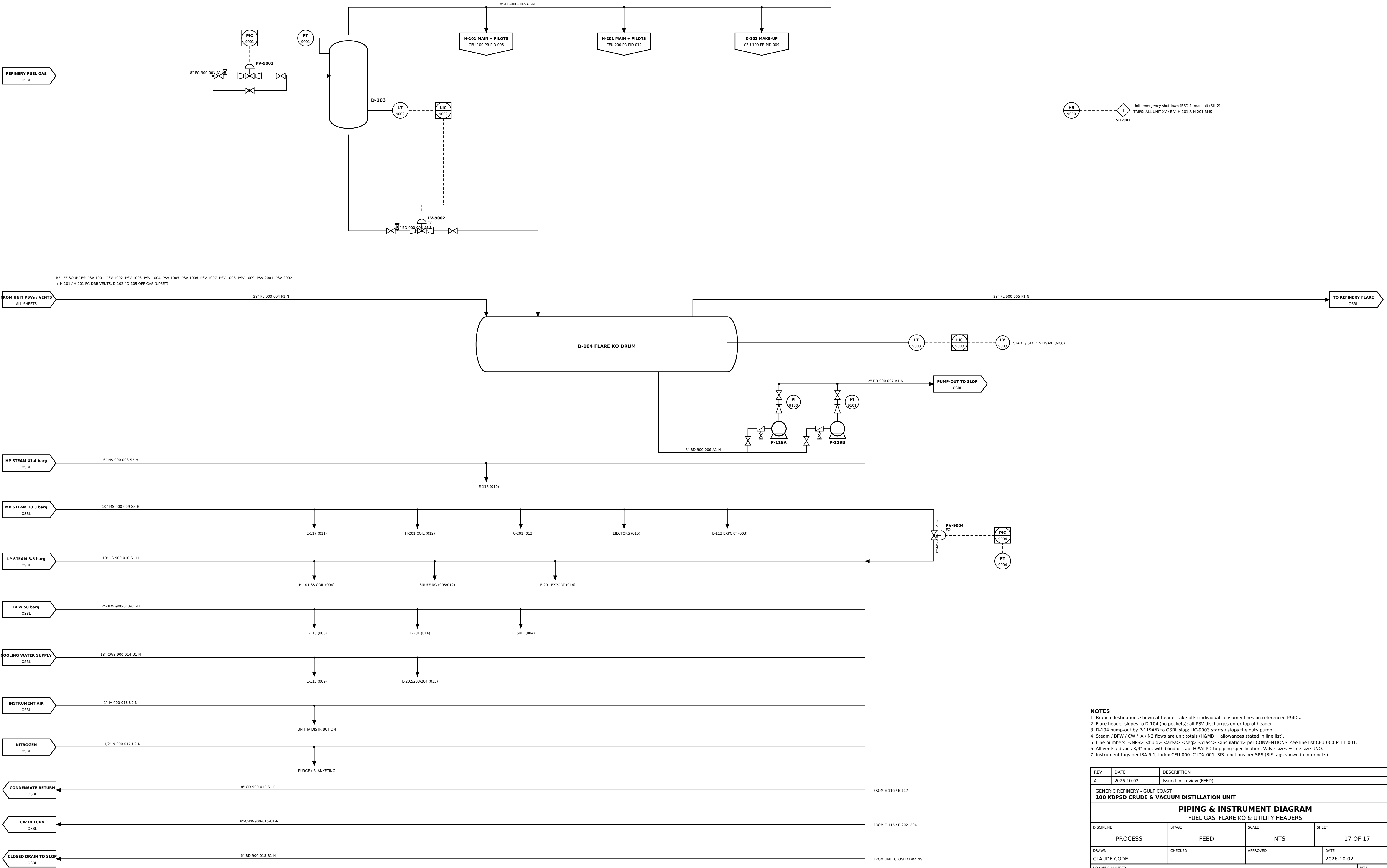
Fuel gas knock-out drum
ID 0.9 m x 2.5 m TTT
DESIGN 7.0 barg / 90 C; CA 3 mm
MAT'L CS

D-104

Flare knock-out drum (unit)
ID 2.1 m x 6.5 m TTT
DESIGN 3.5 barg / 280 C; CA 3 mm
MAT'L CS

P-119A/B

Flare KO drum pump-out
RATED 21 m³/h @ 76 m, 4P 6 bar
MOTOR: 7.5 kW; 4P 682 Phn 11/92
DES. T 90 C; MAT'L S-6 (CS/12Cr)



REV	DATE	DESCRIPTION	
A	2026-10-02	Issued for review (FEED)	
GENERIC REFINERY - GULF COAST 100 KBPSD CRUDE & VACUUM DISTILLATION UNIT			
PIPING & INSTRUMENT DIAGRAM FUEL GAS, FLARE KO & UTILITY HEADERS			
DISCIPLINE	STAGE	SCALE	SHEET
PROCESS	FEED	NTS	17 OF 17
DRAWN	CHECKED	APPROVED	DATE
CLAUDE CODE	-	-	2026-10-02
DRAWING NUMBER			REV
CFU-900-PR-PID-016			A